University of Sharjah

College of Computing and Informatics
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Senior Design Project Report

Phase 1 – Senior Design Project 1

Edge-Cloud Driver Monitoring with Improved User Experience

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Abstract

Nuisance in Driver Monitoring Systems (DMSs) through alerts, whether true or false, leads drivers to disable, circumvent and distrust these systems, exposing drowsy and distracted drivers to a life-threatening risk. The root lies in how these systems must be optimised for User Experience (UX) while still maintaining driver safety. We review previous works and their known limitations, and design a system through the living lens of a tired working person, whose main priority is to reach their destination safely. A webcam feeds a real-time video stream to an object detection model optimised for a Raspberry Pi 5, where pre- and post-processing filter false detections. A confirmed detection cascades downstream into an alert that matches the associated risk, delivered through light, sound and an explained voice message. Confirmed events are logged on the device and later uploaded to a web dashboard that gives the driver tips for safer driving, while no video ever leaves the device. The system is evaluated on a subject-disjoint benchmark, measuring precision, recall and false alarms per hour.

In this first phase, we deliver a complete design rather than a finished product. We specify 26 testable requirements, compare five whole-system alternatives against weighted criteria, and select You Only Look Once (YOLO)11n on a Raspberry Pi 5 with an AI HAT+, after estimating that the CPU alone reaches only about 7 Frames Per Second (FPS). Our four preliminary studies on public data show that YOLO11n leads nano detectors under subject-disjoint testing, that 40% hard-mined negatives cut false positives by 71%, and that cutoffs tuned on some drivers do not hold for others, motivating per-driver calibration. The targets are 25 to 40 FPS, a latency under 100 ms, a per-class recall of at least 0.90 and fewer than 10 false alarms per hour, within a budget of 2725 United Arab Emirates Dirham (AED). We have since built the full software and tested it on a PC, where all 194 automated checks pass and a replay of a held-out driver ran from video to the dashboard over Transport Layer Security (TLS), catching all four phone cues and two of four drinking cues with no detector false alerts. We also recorded four pilot sessions with three of the authors in real cars, by day and by night. Results on the Raspberry Pi, shown on a desk rig in SDP 2, remain for the next phase.

الملخص

تؤدي التنبيهات المزعجة في أنظمة مراقبة السائق، سواء كانت صحيحة أم خاطئة، إلى قيام السائقين بتعطيل هذه الأنظمة أو التحايل عليها وفقدان ثقتهم بها، مما يعرّض السائقين النعسانين والمشتتين لخطر يهدد حياتهم. ويكمن أصل المشكلة في ضرورة تحسين هذه الأنظمة من حيث تجربة المستخدم مع الحفاظ على سلامة السائق. نراجع في هذا التقرير الأعمال السابقة وحدودها المعروفة، ونصمّم نظاماً من منظور شخص عامل متعب، أولويته الوصول إلى وجهته بأمان. تغذّي كاميرا ويب بثاً مباشراً لنموذج لكشف الأجسام محسَّن للعمل على جهاز Raspberry Pi 5، وتقوم مراحل المعالجة القبلية والبعدية بتصفية الاكتشافات الخاطئة. وعند تأكيد أي اكتشاف، يصدر النظام تنبيهاً يتناسب مع مستوى الخطر، ضوئياً وصوتياً مع رسالة صوتية تشرح سببه. تُسجَّل الأحداث المؤكدة على الجهاز ثم تُرفع إلى لوحة تحليلات على الويب تقدّم للسائق نصائح من أجل قيادة أكثر أماناً، دون أن يغادر أي فيديو الجهاز. ويُقيَّم النظام باختبار معياري على أشخاص لم يظهروا في بيانات التدريب، يقيس الدقة والاستدعاء وعدد التنبيهات الخاطئة في الساعة.

في هذه المرحلة الأولى نقدّم تصميماً متكاملاً وليس منتجاً نهائياً. فقد حدّدنا ٢٦ متطلباً قابلاً للاختبار، وقارنّا خمسة بدائل للنظام بأكمله وفق معايير موزونة، واخترنا نموذج YOLO11n على Raspberry Pi 5 مع مسرّع AI HAT+، بعد أن قدّرنا أن المعالج المركزي وحده لا يتجاوز نحو ٧ إطارات في الثانية. وتُظهر دراساتنا الأولية الأربع على بيانات عامة أن YOLO11n يتفوّق على النماذج الصغيرة الأخرى عند الاختبار على سائقين جدد، وأن إضافة ٤٠٪ من العينات السلبية الصعبة تخفض الاكتشافات الخاطئة بنسبة ٧١٪، وأن عتبات القرار المضبوطة على بعض السائقين لا تصلح لغيرهم، مما يبرّر المعايرة لكل سائق. وتشمل الأهداف ٢٥ إلى ٤٠ إطاراً في الثانية، وزمن استجابة أقل من ١٠٠ ميلي ثانية، واستدعاءً لا يقل عن ٠,٩٠ لكل فئة، وأقل من ١٠ تنبيهات خاطئة في الساعة، ضمن ميزانية تبلغ ٢٧٢٥ درهماً. وقد بنينا البرمجيات كاملةً واختبرناها على حاسوب، فنجحت جميع الفحوصات الآلية البالغ عددها ١٩٤، ورصدت إعادة تشغيل تسجيل لسائق لم يتدرب عليه النموذج، من الفيديو حتى لوحة المعلومات عبر اتصال مشفّر، جميع حالات استخدام الهاتف الأربع وحالتين من أربع حالات شرب، دون أي إنذار خاطئ من الكاشف. كما سجّلنا أربع جلسات تجريبية لثلاثة من مؤلفي التقرير داخل سيارات حقيقية، نهاراً وليلاً. وتبقى نتائج Raspberry Pi، التي ستُعرض على منصة مكتبية في المرحلة الثانية، للمرحلة التالية.

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Symbols, Keywords, and Abbreviations

AC	Air Conditioning. 6
ACM	Association for Computing Machinery. 19
AED	United Arab Emirates Dirham. iii, 10, 12, 20, 22, 23, 43, 54, 55
AP	Average Precision. 15
API	Application Programming Interface. 23, 46, 56, 57, 59
CNN	Convolutional Neural Network. 8–12
COCO	Common Objects in Context. 2, 10, 13, 30, 32
DMS	Driver Monitoring System. iii, 1, 2, 6, 7, 65
EAR	Eye Aspect Ratio. 5, 8, 9, 16, 19, 21, 34, 35, 38, 50, 59, 61, 69
EMA	Exponential Moving Average. 4, 9, 14, 23, 24, 27, 38, 40, 42, 53, 56, 59, 62, 63, 65
EU	European Union. 1, 8, 44
FLOP	Float Point Operation. 5
FN	False Negative. 15
FP	False Positive. 5, 15
FPS	Frames Per Second. iii, 2, 4, 13, 14, 16, 19–22, 24, 33, 40, 46–50, 52–54, 58, 59, 61, 64
GDPR	General Data Protection Regulation. 17, 42
GPIO	General-Purpose Input/Output. 25, 46
GPS	Global Positioning System. 3, 38, 69

GPU	Graphics Processing Unit. 9, 40
GSR	General Safety Regulation. 1, 8, 44
HDMI	High-Definition Multimedia Interface. 23, 25
Hz	Hertz. 33
IEEE	Institute of Electrical and Electronics Engineers. 19
INT8	8-bit Integer. 13, 15, 24, 31, 32, 41, 42, 61, 63
IoU	Intersection over Union. 13, 15, 50
IR	Infra-Red. 1, 6, 10, 11, 22, 48, 66
JSON	JavaScript Object Notation. 35
LCD	Liquid Crystal Display. 23
LED	Light-Emitting Diode. 4, 5, 17, 19, 23, 44
mAP	Mean Average Precision. 13, 15, 22, 31, 32, 40–42, 52, 61
MATLAB	Matrix Laboratory. 33, 46, 59
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor. 23, 25, 29, 33, 54, 55, 69
NIR	Near-Infrared. 8, 10, 11, 22, 43
NMS	Non-Maximum Suppression. 13
NPU	Neural Processing Unit. 10, 13, 15, 21–24, 27, 29, 30, 42–44, 52, 60, 62
ONNX	Open Neural Network Exchange. 13, 15, 32, 58, 59, 61, 63
PCIe	Peripheral Component Interconnect Express. 13, 25
PDPL	Personal Data Protection Law. 17, 19, 42
PERCLOS	Percentage of Eye Closure. 5, 16, 19, 33, 38, 59, 61

PIN Personal Identification Number. 20, 24, 25, 38, 59, 69

PoI Point of Interest. 5, 53, 57

PTQ Post-Training Quantization. 15, 32, 42, 46, 52

PWM Pulse-Width Modulation. 23

QAT Quantization-Aware Training. 15, 42, 46, 52, 61, 63

RGB Red Green Blue. 2, 4, 6, 10, 11, 20, 22–24, 29, 44, 47, 48, 55, 66

RoI Region of Interest. 3, 6, 32, 42

RTA Roads and Transport Authority. 3, 44

SDP Senior Design Project. 3, 7

TLS Transport Layer Security. iii, 3, 5, 17, 20, 23–26, 35, 59, 61, 63

TOPS Tera Operations Per Second. 13

TP True Positive. 15

UAE United Arab Emirates. 1, 10–12, 17–20, 22, 42–44, 54

UI User Interface. 5, 20, 56

USB Universal Serial Bus. 4, 13, 17, 22, 23, 25, 27, 29, 46, 55

UVC USB Video Class. 17, 22

UX User Experience. iii, 1, 5, 7

WAL Write-Ahead Logging. 5, 17, 20, 24, 34, 59

WCAG Web Content Accessibility Guidelines. 17

YOLO You Only Look Once. iii, 2, 4, 13, 21, 22, 24, 27, 30, 32, 40–42, 46, 57, 58, 65, 66

Chapter 1 Introduction

Problem Statement

In this work, we propose a plan for the design and implementation of an edge-cloud DMS with the focus on UX. This solves a critical deployment issue which is the nuisance caused by driver safety systems. Distrust caused by alerts—whether be true or false alerts—exposes a life-threatening risk on drowsy and distracted drivers. Our design combines:

- UX through multi-tier risk-specific explainable alerts, web dashboard analytics and multi-modal alerts.
- Privacy by processing all video data on-device, only text is logged and uploaded to the cloud.
- Extended safety and performance through data curation, model compression and system optimization.

The system is embedded onto a Raspberry Pi 5, a live camera feeds the model, processing filters false detections and a confirmed event cascades downstream into suitable alerts.

1.1 Motivation

DMSs evolved since they got introduced in 2006 [1]. Transitioning from merely alerting the driver through an Infra-Red (IR) camera and applying breaks [2], to tracking eye gaze and fusing it with vehicular spatial awareness [3]. Conversely, nuisance alerts and false alarms causes driver distrust in these systems, causing system neglect or circumvention [4]. The gap falls under improving the user experience in theses systems. Given the United Arab Emirates (UAE) context of high speed highways, and the need for private and trustworthy systems. We aim to deliver a system that is built to supplement the driver's attention and not replace it. Figure 1.1 shows the results of in-attentive driving, which stresses the need for these systems to be deployed at a greater rate. In 2024, European Union (EU)'s General Safety Regulation (GSR) mandates that all newly registered vehicles must include a DMS [5]. To supplement this surging rate of deployment, the need for systems that

allows the user to entrust them becomes apparent. We aim to achieve this by adaptive and enhanced alerting, driving analytics and a reliable drowsiness and distraction monitoring system.

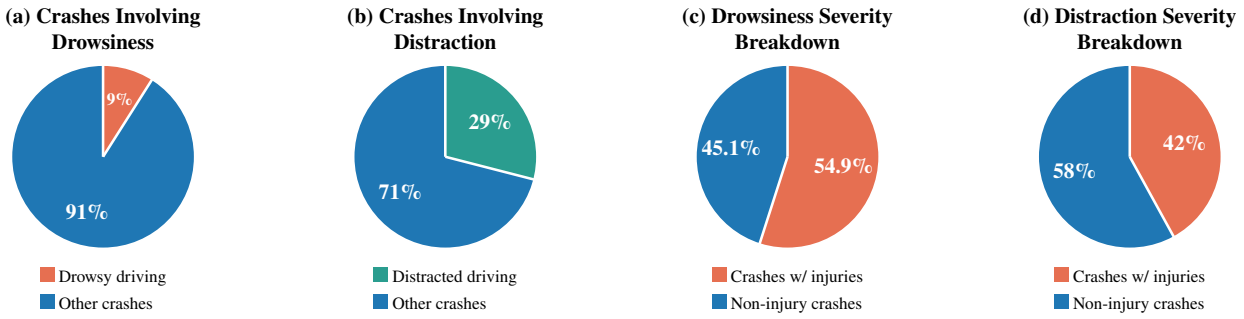


Figure 1.1: (a) Drowsy driving involvement in all crashes [6]; (b) Distracted driving involvement in all crashes [7]; (c) Severity of drowsiness-caused crashes [8]; (d) Severity of distraction-caused crashes [9]

1.2 Limitations of Traditional Systems

DMS is notorious for the high rate of alerts, a study argues that frequent or overlapping alerts can negatively affect driver mental models.Furthermore, the authors stress a need for improved contextual calibration in future driver monitoring systems designs [10]. Figure 1.2 highlights a local survey where drivers relay the common causes of distraction. This exposes a common pitfall that traditional DMS suffers from. By only tracking the driver's face and eyes, these systems cannot detect external distractions like road signs and traffic, nor can they detect noise and mental distractions caused by passengers or car controls. Later in this report, we discuss further details on how we effectively counter the aforementioned shortcomings, using initial setup driver calibration, a user-friendly system configuration device, and context-aware alerting.

1.3 Key Differentiators

Our solution has several clear differentiators from previously deployed systems. The overall system is built on a lightweight YOLO object detector that is further pruned and quantized to run at a streamlined latency with high FPS throughput. The model is initially pre-trained via Common Objects in Context (COCO). Then, fine-tuned via our own subject-disjoint in-cabin Red Green Blue (RGB) video dataset, which involves ranging lighting conditions to train a model where it is most critical. Later, we extend the dataset to include harder negatives to help reduce false alarms. We pair this detection model with a casual temporal filter to reduce random detection spikes and improve user experience via escalating alarms based on risk assessment. The system extends these

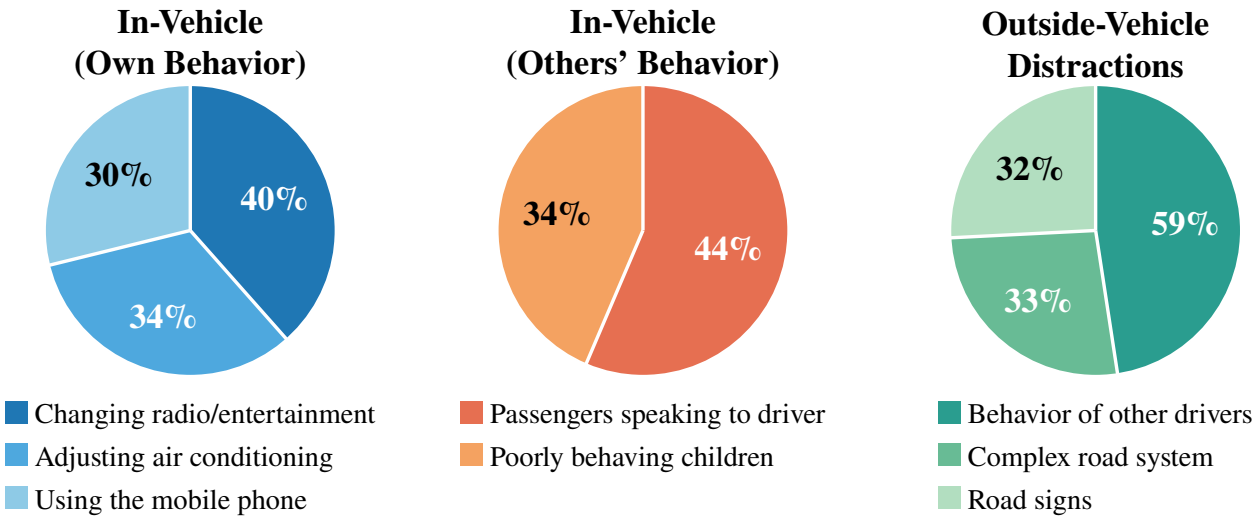


Figure 1.2: (a-c) Local survey on driver distraction causes [11]

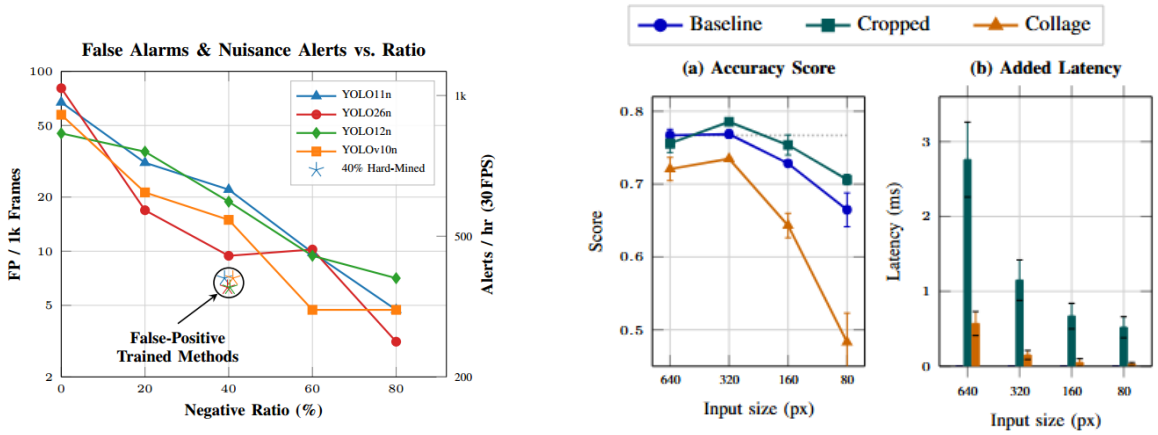
with a privacy-first logging system where only event records are transmitted via TLS to a server database where a web dashboard relays driving analytics, giving feedback via tips and behavior trends to the driver after the session. A driver-facing easy-to-navigate tablet allows for system calibration and initial setup. We pair traditional alerts with intelligent voice messages in English and Arabic for further transparency and rest-stop location suggestions via a Global Positioning System (GPS).

Aside from these differentiators, the team initially conducted research via empirical methods to evaluate several object detection models, and the effect of hard-negative mining on false alarms. We later conducted ablation studies on temporal filters and the effect of ranging input resolution and Region of Interest (RoI) cropping on input video streams, which led to most of our design choices in this report to be backed by real evaluations.

We aim to provide this trustworthy solution to everyone. Primarily, our solution serves both the general public and commercial vehicle fleet operators. Specifically, government agencies such as Dubai's Roads and Transport Authority (RTA), which deploys thousands of vehicles and employs personnel who drive for extended periods (such as taxi drivers), would benefit from a private, dependable safety system to support these operators.

1.4 Scope and Feature Priorities

Features are grouped by priority, "Required" features that deliver the base build, "Desired" features that will be delivered depending on the remaining time budget, and "Future" features that are out of scope of our Senior Design Project (SDP) 2. In the following sections we list the breakdown in detail.



(a) Hard-negative mining

(b) RoI cropping methods

Figure 1.3: Empirical evaluation conducted in this work (**Ours**):
(a) Suppression via hard-negatives ; and (b) Evaluating RoI cropping strategies.

Table 1.1: Required System Feature Specifications

ID	**Title**	**Description**
FS-01	Deployment on Raspberry Pi 5	The full detection pipeline runs live on the Pi 5.
FS-02	1080p 30 FPS RGB camera	A Universal Serial Bus (USB) 3.0 webcam aimed at the driver's face records 1080p at 30 FPS. Each frame is cropped to a fixed region before detection.
FS-03	Object detection model	YOLO11n detects drowsy and distraction classes. It is fine-tuned on the subject-disjoint dataset below.
FS-04	Temporal filtering (Exponential Moving Average (EMA))	An exponential moving average over each class score suppresses random single-frame spikes. It is defined in seconds, not frames, so it behaves the same at any FPS.
FS-05	Speaker and Light-Emitting Diode (LED) bar alert	After a confirmed detection, a bright LED bar flashes and a USB 3.0 speaker plays a voice warning.
FS-06	Alert cool-down	Cheap and effective suppression for repeated warnings for the same event within a time frame.
FS-07	Mirror-check suppression	The classes (by design) do not depend on head orientation, so normal mirror checks and head turns do not trigger alerts.
FS-08	Internal logging	Each confirmed detection (time, type, confidence) is saved to the Pi's local storage. No raw video is kept by default.
FS-09	On-device video processing	All video processing happens on the Pi and raw video never leaves the vehicle.
FS-10	Subject-disjoint dataset collection	10-minute RGB in-cabin videos over several sessions, recorded in the afternoon and evening. Subjects are split so no subject appears in more than one of train, val, and test.

Table 1.2: Desired System Feature Specifications

ID	Title	Description
FS-11	Model pruning and quantization	Pruning removes channels to cut Float Point Operations (FLOPs). Quantization methods are compared on the Pi, choosing the best FPS while retaining detection metrics.
FS-12	Hard-negative dataset	Subjects are re-recorded performing look-alike non-violations flagged by the baseline model. Only negative frames are added.
FS-13	Neutral baseline clips	Each subject records a 20 s neutral baseline clip to benchmark and evaluate calibration.
FS-14	Class-specific alerting	Distinct alert tiers and voice notifications per class.
FS-15	Tiered alert escalation	Gentle notification → voice prompt → multi-modal alarm (LED bar and voice) as risk persists.
FS-16	Clear, explainable alerts	Contextual alerts (e.g., "yawning detected 3 times in the last 5 minutes") rather than generic warnings to build driver trust.
FS-17	Camera-blind check	Measures frame brightness and sharpness. Logs and issues "camera blocked" warning if unclear for ≥ 3 s.
FS-18	Driver-centered crop tracking	Lightweight face detector runs at $\sim$ 1 Hz on lower resolution frames to dynamically adjust the face crop.
FS-19	New-driver calibration	Driver sits neutrally for 20 s while detector runs at low confidence. Cutoffs set to 99th percentile + a safe gap. Increasing the confidence threshold and reducing false alarms
FS-20	Eye closure and micro-sleep alert	Face landmark tracking calculates Eye Aspect Ratio (EAR), confirmed by a blink score and head pitch. Continuous closure triggers loud alert; 60 s Percentage of Eye Closure (PERCLOS) triggers gentle fatigue prompt.
FS-21	Power-cut-safe logging	Database operates in Write-Ahead Logging (WAL) mode to prevent corruption during power interruptions.
FS-22	Dashboard and cloud sync	Post-trip event records uploaded over TLS to update database and web dashboard with trends and metrics.
FS-23	Actionable post-trip tips	Recommendations generated from logged behaviors (e.g., "take a 30 minute break")
FS-24	Driver-facing tablet	Dedicated low-distraction glance-able User Interface (UI), allows for easy system calibration.
FS-25	Pre-trip readiness check	Pre-drive fatigue survey on tablet recommending rest and nearby rest locations if fatigue is reported.
FS-26	Rest-stop suggestions	Queries the phone's location and Point of Interest (PoI) map services to recommend nearby rest stops or fuel stations when violations exceed a threshold.

1.5 Proposal Revisions

Our initial proposal covered the pre-development in detail, discussing a methodology for evaluating different object detectors and studying the effect of hard-negative mining. The primary focus was edge system performance on accuracy metrics and False Positives (FPs) per 1000 frames. Since then, we changed the deployment hardware, upgrading from a Raspberry Pi 4 to a Pi 5 to provide better throughput and handle larger systems. We then pivoted the project to provide better UX

Table 1.3: Future System Feature Specifications

Status	Title	Description
Future	Driving-context suppression	Suppress or defer warnings during reversing maneuvers, standstill red lights, or high-density traffic.
Future	Multi-modal fusion	Aggregate multi-spectral video streams.
Future	Multiple camera feed	Additional camera to capture wide-angle context or secondary angles.
Future	Smartwatch and wearables integration	Incorporate driver heart-rate or biometric motion telemetry to augment edge detection.
Future	Robustness study	Systematic stress testing across glare, low light, optical blur, pose deviations, eyewear (glasses/sunglasses), face coverings, day-to-night transfer, and multi-vehicle cabins.
Future	Adaptive sensitivity by context	Context-aware threshold modification depending on the driver's behavior patterns.
Future	Steering wheel vibration	Haptic vibration on the steering wheel to reduce driver reaction latency.
Future	Cabin comfort integration	System can change the Air Conditioning (AC) or change the volume of what's playing via the car's speakers.
Future	Voice recognition	Driver audio-interface enabling inquiries (e.g., "How tired am I?").

through various adaptive alerting mechanisms, nuisance filters, and system components such as web analytics or a user-friendly tablet interface.

1.6 Project Roadmap

We studied and evaluated different object detectors on RGB and IR public DMS datasets. Followed by evaluating negative-mining on false detections, and an empirical study on the trade-offs between temporal filters. Finally, we studied the effect of pruning, RoI cropping and input resolutions on performance.

We will establish our organic in-cabin video dataset, with an extension of harder non-violation clips that will harden our detection model. Delivering a public subject-disjoint dataset.

Before deployment, we perform an extensive literature review, to find what do drivers suffer from by using traditional DMS. Aiming to effectively counter these pain-points through careful system planning and design. we plan to document every design change meticulously. If the system faces technical complexity issues, we plan to surgically remove any non-required component, ensuring project delivery on time.

Post-deployment, we evaluate the system by involving users for their feedback, adequate testing and taking recommendations from peers, allowing the project to develop and adapt to fit the end user the best.

1.7 Report Organization

To end this chapter, here is a breakdown on how this report is organized:

In Chapter II, we explore current limitations and shortcomings in existing DMS work, providing a gap summary highlighting the need for better UX in said systems.

In Chapter III, we deep-dive into the technical concepts and principles of deploying complex DMSs.

In Chapter IV, we report in detail our proposed solution, funneling a general idea into a detailed look at every component in the system. To provide a complete report, project constraints and issues are discussed.

Chapter V includes the methodology, highlighting the empirical methods we used to evaluate, organize and build the system and its dataset.

Chapter VI reports what has been implemented so far, the measurements taken on a PC and the state of every specification.

Finally Chapter VII summarizes the main findings, lists the project milestones and discusses what we aim to work on in SDP 2.

Chapter 2 Literature Review

In this chapter, we review previous works in academic detection, research on alert design and driver trust, commercial products and public datasets, concluding with the gaps our system is designed to fill.

2.1 Academic research on driver monitoring

The majority of distraction research classifies a whole frame with a Convolutional Neural Network (CNN), reaching about 96% accuracy on the AUC dataset [12, 13]. Once the test drivers are unseen, accuracy falls to about 90% [14], revealing how much of the earlier gain came from frames of the same person appearing in both splits. Drowsiness research mostly relies on facial landmarks such as the EAR [15, 16, 17], or on models of fatigue over time [18]. These methods are lightweight, yet they fail exactly when a hand, a bottle or a phone hides the face. As Table 2.1 shows, most studies evaluate on drivers the model has already seen, few are deployed on embedded hardware, none reports false alarms per hour, and none calibrates its thresholds per driver.

2.2 Alert design, trust and user experience

Detection is only half of the problem, since the alert decides whether the driver keeps the system switched on. Frequent or unjustified alerts annoy drivers, eroding their trust and their willingness to respond [20, 21, 22, 4]. Explaining what the system does and why it alerted raises trust [23, 24, 25, 26]. Graded warnings outperform single-stage ones [27, 28, 29], and while audio is the fastest modality, it can startle the driver, performing best when combined with light or haptics [30, 31, 32, 33, 34]. Table 2.2 groups these findings with the design decision each one drives.

2.3 Commercial driver monitoring systems

Factory-fitted systems such as Smart Eye [39] use a hidden Near-Infrared (NIR) camera and are tuned to the EU GSR [5]. Fleet products such as Seeing Machines Guardian [40], Lytx DriveCam

Table 2.1: Summary of academic driver monitoring studies

Ref	Task	Method	Dataset	Metrics	Hardware	Limitation
[12]	Distraction (10 postures)	Genetically weighted ensemble of AlexNet and Inception	AUC v1	95.98% accuracy	Desktop Graphics Processing Unit (GPU)	Frame classifier, heavy ensemble, no deployment
[13]	Distraction (10 postures)	Modified VGG-16	AUC v1	96.31% accuracy	Desktop GPU	Very large model, single frame, no alert logic
[14]	Distraction (10 postures)	CNN ensemble on face, hands and skin regions	AUC v2 (driver split)	90% accuracy	Desktop GPU	Accuracy drops once drivers are split, not embedded
[16]	Drowsiness	Facial landmarks and a multilayer perceptron	NTHU-DDD	Above 80% accuracy	Android phone	Landmarks lost under hand or object occlusion
[17]	Drowsiness	Face tracking, eye and mouth openness rules	Own videos	Accuracy	PC	Fixed thresholds, no per-driver calibration
[18]	Drowsiness	Hierarchical temporal deep belief network	NTHU-DDD	Accuracy, F1	Desktop GPU	Acted clips, no deployment, no alert design
[15]	Blinks	EAR from six landmarks and an SVM	ZJU, Eyeblink8	Precision, recall	CPU, real time	Eyes only, sensitive to head pose and occlusion
[19]	Yawning	Face and mouth detection baseline	YawDD	Detection rate	PC	Yawning only, no covered yawns or distraction

Table 2.2: User experience findings and the design decision each one drives

Theme	Studies	Finding	Our design response
Nuisance and false alarms	[20, 21, 22, 4]	Frequent or unjustified alerts cause annoyance, distrust and ignored warnings	EMA filter, confirmation time, cool-down, hard negatives, false alarms per hour as a target
Transparency	[23, 24, 25, 26]	Explaining what the system does and why it alerted raises trust and willingness to use	Explainable voice messages, onboarding on the display, post-trip dashboard
Graded alerts	[27, 28, 29]	Multi-stage warnings shorten response time and are preferred over single-stage ones	Three-tier escalation per class
Modality	[30, 31, 32, 33, 34, 35]	Audio is fast but can startle, haptic helps under visual load, and combining modalities reduces missed warnings	Voice plus peripheral light, haptic wheel prototype, gentle first tier
Drowsy drivers	[36, 37]	Warnings help drowsy drivers mostly when they look away, and stimuli must be tuned for fatigue	Separate drowsiness tones, eye-closure tier, rest-stop suggestions
Personalisation	[25, 38]	Adapting to the individual driver improves perceived intelligence and experience	Per-driver calibration of cut-offs

[41] and Samsara [42] send events and video to a cloud platform, where a safety manager reviews them. These products require a subscription or a new vehicle, upload driver footage, and none of them publishes false alarms per hour or exposes a per-driver calibration to the driver (Table 2.3).

Table 2.3: Commercial driver monitoring products

Product	**Sensing**	**Features**	**Cloud needed?**	**Price**
Seeing Machines Guardian [40]	IR driver camera with in-cab alarm	Fatigue and distraction detection, seat vibration and audio, events reviewed by a monitoring centre	Yes	Quote and subscription
Smart Eye [39]	NIR camera, factory or aftermarket	Gaze, drowsiness, distraction and driver identification for car makers and fleets	Optional	Built into new cars or quoted
Lytx DriveCam [41]	Dual-facing video recorder	Risky behaviour detection, video events, coaching workflow	Yes	Subscription
Samsara AI dash cam [42]	Dual-facing camera	Phone use and drowsiness detection, in-cab audio alerts, fleet dashboard	Yes	Subscription
This project	RGB webcam, on-device Neural Processing Unit (NPU)	Explained tiered alerts, calibration, event-only analytics	Only after the trip	About 2725 AED prototype

2.4 Public datasets and why we collect our own

No public dataset satisfies our needs on its own. Our four classes are spread across datasets with different camera views and label definitions, hand-over-mouth is rarely labelled as its own class, and none contains hard negatives or the sun and glare of UAE cabins (Table 2.4). We therefore pre-train on COCO [43], use the public datasets for the studies in Chapter 1, and collect our own subject-disjoint dataset (Section 5.3).

2.5 Comparison of approaches and the gap this project fills

Table 2.5 groups the reviewed work into families. Landmark rules run on-device but cannot see objects, CNN classifiers recognise objects but are heavy and rarely tested on unseen drivers, and commercial systems are robust but closed or cloud-dependent. No family satisfies all six needs in the last row, and each need maps to a feature of Section 1.4.

Table 2.4: Public datasets and their gaps for this project

Dataset	**Classes**	**Camera view**	**Subject split?**	**Gap for us**
State Farm Distracted Driver [44]	Ten postures incl. texting, phone call, drinking	Side view, still images	Hidden test set, often split at random	No yawning, wrong view, no video
YawDD [19]	Normal, talking/singing, yawning	Mirror and dashboard	Not defined	No distraction classes, few covered yawns
NTHU-DDD [18]	Drowsy cues: yawning, nodding, slow blinks	Frontal, RGB and IR	By subject	Acted, no phone or drinking
DMD [45]	Distraction, drowsiness, gaze, hands	Face, body and hands, RGB/IR/depth	Possible by subject	Different label set, no UAE conditions, no hard negatives
Drive&Act [46]	Fine-grained in-cabin activities	Multiple NIR and depth views	By subject	Automated-driving activities, NIR only

Table 2.5: Gap table across approach families

Approach / work	**On-device?**	**Hand-over-mouth, drinking, phone?**	**Subject-disjoint evaluation?**	**Reports false alarms/h?**	**Per-driver calibration?**	**No raw video?**
Landmark / rule-based [15, 16, 17]	Yes	No	Rarely	No	No	Yes
CNN classification [12, 13, 14]	No	Partly	Rarely	No	No	Yes
Object detection (academic)	Rarely	Yes	Rarely	No	No	Yes
Commercial IR [39]	Yes	Partly	Not public	Not public	No	Yes
Cloud-based fleet [40, 41, 42]	Partly	Partly	Not public	Not public	No	No
This project	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

Chapter 3 Technical Background

3.1 Core concepts

3.1.1 Driver distraction and drowsiness: what the system must recognise

Distraction is commonly divided into three types [47, 48]. Visual distraction takes the eyes off the road, manual distraction takes a hand off the wheel, and cognitive distraction takes the mind off driving, with phone use being the clearest example as it combines all three. Drowsiness behaves differently, building up gradually, appearing first as yawning and rubbing of the face, then as longer blinks, and finally as micro-sleeps where the eyes remain closed for a second or more.

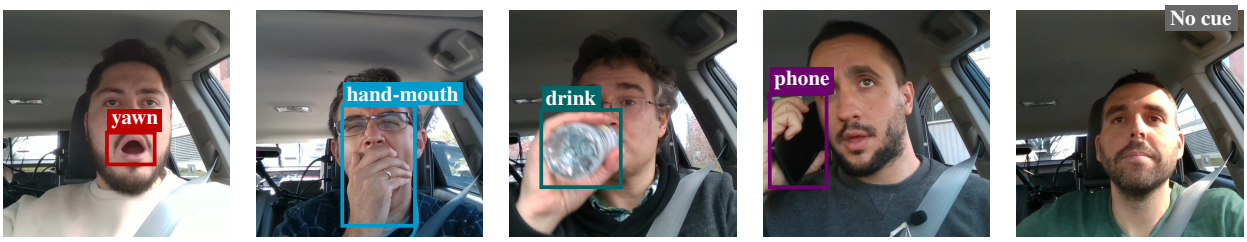
We chose four visible cues that a camera can capture reliably (Table 3.1 and Figure 3.1). Yawning and hand-over-mouth, which is usually a covered yawn, are early signs of fatigue. Drinking and phone use are manual and visual distractions, and phone use alone carries an 800 AED fine and four black points in the UAE [49]. None of these classes depends on where the head points, ensuring that ordinary mirror checks do not trigger alerts. Eye closure is handled separately through a landmark module (Section 3.2.4).

Table 3.1: Target classes and what each one indicates

Class	**Group**	**What it indicates**	**Why a detector is needed**
Yawning	Drowsiness	Early fatigue, often repeated before micro-sleeps	Mouth shape is clear, but often partly covered
Hand-over-mouth	Drowsiness	A covered yawn or rubbing the face	Hides the mouth landmarks that rule-based methods use
Drinking	Distraction	Manual and visual distraction	Needs the bottle or cup to be seen as an object
Phone use	Distraction	Manual, visual and cognitive distraction	Needs the phone near the ear or in the hand

3.1.2 Object detection with CNNs and the YOLO family

A CNN learns small filters that slide over an image and respond to edges, textures and shapes. Stacking many of these layers forms a backbone that turns the image into feature maps at several



Yawning | Hand-over-mouth | Drinking | Phone use | Negative

Figure 3.1: The four target classes with their boxes and a negative frame, on DMD frames [45] (from [50]). A covered yawn is labelled hand-over-mouth

scales. A neck then mixes these scales so that both small objects, such as a phone at the ear, and large ones, such as an open mouth, are represented, and a head predicts a bounding box, a confidence score and a class for every location. Since many neighbouring locations predict the same object, Non-Maximum Suppression (NMS) keeps the highest-scoring box and removes the others that overlap it above an Intersection over Union (IoU) threshold. A one-stage detector performs all of this in a single pass, which makes it fast enough for video [51]. Figure 3.2 illustrates this structure.

YOLO11n is the smallest variant of the Ultralytics YOLO11 family [52, 53], with 2.6 million parameters and 6.5 GFLOPs per 640×640 image while reaching 39.5 Mean Average Precision $(\text{mAP})_{50\text{–}95}$ on COCO. We rely on transfer learning: the network starts from weights trained on COCO [43], so it already recognises hands, cups and phones, and only needs to learn our four classes from a much smaller dataset.

3.1.3 Running AI at the edge: Raspberry Pi 5 and the AI HAT+

The Raspberry Pi 5 has a quad-core Arm Cortex-A76 at 2.4 GHz, two USB 3.0 ports and a Peripheral Component Interconnect Express (PCIe) 2.0 ×1 lane, and requires a 5 V, 5 A supply [54]. On the CPU alone, Ultralytics reports YOLO11n at 6.79 FPS through Open Neural Network Exchange (ONNX) [55], far below our target. The AI HAT+ adds a Hailo NPU on the PCIe lane, rated at 13 or 26 Tera Operations Per Second (TOPS) depending on the variant [56], running an 8-bit Integer (INT8) model compiled with the Hailo toolchain [57]. Since the USB ports are limited to 600 mA on weaker supplies and to 1.6 A with a 5 A supply, power planning matters. Heavy load also heats the board, which throttles above 80–85 °C, making the active cooler a requirement. Processing everything on-device means that raw video never needs to leave the device, and only event records do.

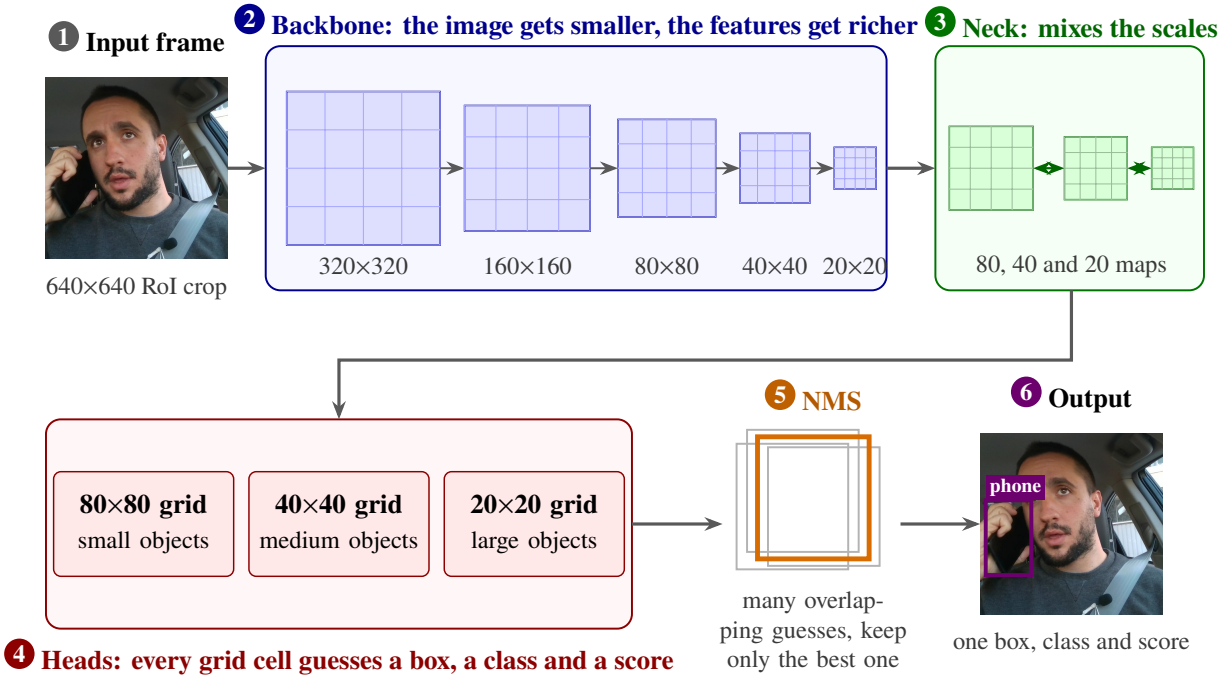


Figure 3.2: How YOLO11n turns one frame into one box: the backbone shrinks the image into feature maps, the neck mixes fine and coarse maps, three heads guess boxes for small, medium and large objects, and NMS keeps the best box

3.2 Principles used in the project

3.2.1 Temporal smoothing with an exponential moving average

A detector scores every frame independently, so a single odd frame can produce a spike. We therefore smooth each class score x_t with an EMA:

$$s_t = \alpha\, x_t + (1 - \alpha)\, s_{t-1}, \qquad \alpha = 1 - e^{-\Delta t/\tau} \tag{3.1}$$

where Δt is the measured time since the last frame and τ is a time constant in seconds. Defining α through Δt ensures the filter forgets at the same rate in seconds whether the pipeline runs at 15 or 30 FPS, as Figure 3.3 shows. A larger τ suppresses more noise at the cost of a slower response: for a cue that suddenly appears, the smoothed score crosses a cutoff θ after

$$t_\theta = -\tau \ln(1 - \theta). \tag{3.2}$$

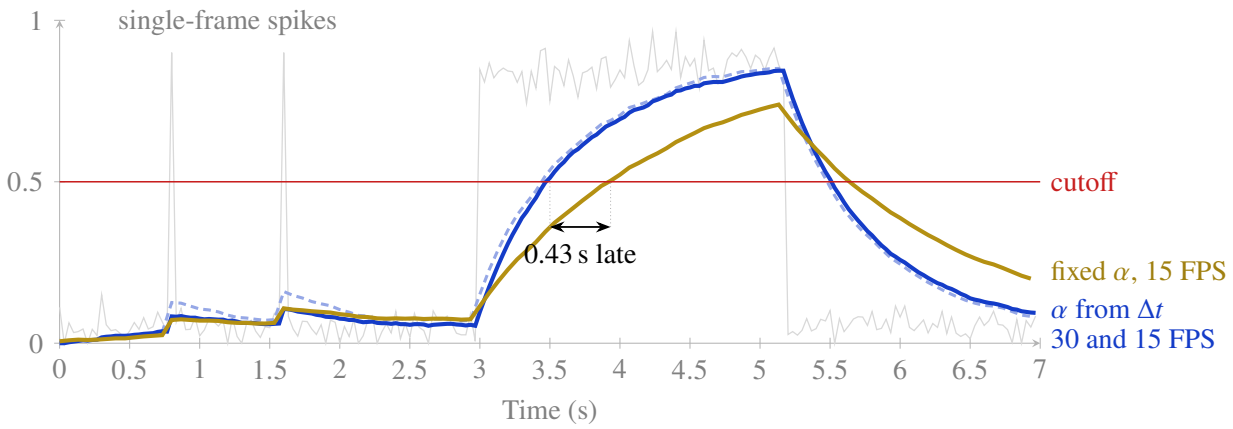


Figure 3.3: The same cue filtered at 30 and 15 FPS with $\tau = 0.6$ s: both cross the cutoff at almost the same time, single-frame spikes stay far below it, and a fixed per-frame α reused at 15 FPS reacts about 0.43 s late (illustrative signal)

3.2.2 Model compression: pruning and quantization

Models can be made smaller and faster in two ways. Structured pruning removes whole filters or channels whose weights contribute little, usually judged by their magnitude [58]. Unlike removing single weights, this shrinks the actual tensors, so the CPU or NPU performs less work without special sparse kernels. The model is then fine-tuned again to recover the lost accuracy.

Quantization stores weights and activations with fewer bits. FP16 halves the memory with almost no loss in accuracy, while INT8 maps every value to one of 256 levels using a scale s and a zero point z, $q = \text{round}(x/s) + z$ [59]. Post-Training Quantization (PTQ) selects s and z from a small calibration set after training, whereas Quantization-Aware Training (QAT) simulates the rounding during training and usually loses less accuracy. On the Pi, runtimes such as ONNX Runtime, NCNN [60] and TFLite run on the CPU, while the Hailo Dataflow Compiler quantizes the model to INT8 and compiles it for the NPU [57].

3.2.3 Measuring detector performance

A detection counts as a True Positive (TP) when its class is correct and its IoU with a labelled box is at least 0.5. Otherwise, it is a FP, and a missed labelled box is a False Negative (FN). From these counts:

$$P = \frac{TP}{TP + FP}, \qquad R = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2PR}{P + R}, \qquad \text{IoU} = \frac{|A \cap B|}{|A \cup B|}. \tag{3.3}$$

Average Precision (AP) is the area under the precision-recall curve of one class, and mAP averages it over classes, either at an IoU of 0.5 or across thresholds from 0.5 to 0.95. For the running system,

we add throughput in FPS, end-to-end latency, and the metric that matters most to a driver:

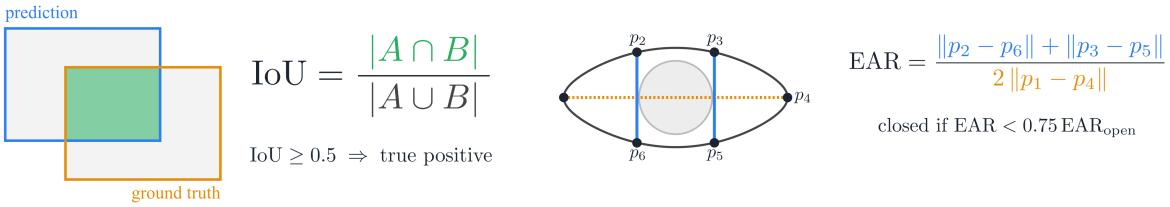
$$\text{False alarms per hour} = \frac{\text{confirmed alerts on clips with no true cue}}{\text{clip duration in hours}}. \tag{3.4}$$

3.2.4 Eye-closure measures: EAR and PERCLOS

The EAR of Soukupová and Čech [15] uses six eye landmarks p_1 to p_6:

$$\text{EAR} = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\,\|p_1 - p_4\|}. \tag{3.5}$$

The EAR stays roughly constant while the eye is open and drops towards zero as it closes. Since eye shape varies between people, a fixed threshold does not suit every driver, so we take each driver's open-eye value from a 20 s baseline and treat the eye as closed below 50% of it. The EAR also drops when a driver looks down, so it cannot be used on its own. PERCLOS is the fraction of time the eyes are closed within a sliding window, here 60 s, and is a validated measure of drowsiness [61]. Figure 3.4b labels the landmarks.



(a) IoU between a predicted and a ground-truth box (b) Eye landmarks p_1–p_6 used for the EAR

Figure 3.4: The two geometric measures used in this project

3.3 Relevant standards, protocols and interfaces

The system uses common interfaces and follows the standards in Table 3.2.

Table 3.2: Standards, protocols and how we apply them

Standard / protocol	**What it governs**	**How we apply it**
USB / USB Video Class (UVC) [62]	Plug-and-play video devices	Driverless capture of the C922 through V4L2
TLS 1.3 [63]	Encrypted, authenticated transport	All uploads and map queries use HTTPS
SQLite WAL [64]	Crash-safe local database	Event log survives power cuts at ignition-off
Web Content Accessibility Guidelines (WCAG) 2.1, SC 2.3.1 [65]	No more than three flashes per second	LED patterns capped at 3 Hz in software
ISO 16750-2 [66]	Supply voltage ranges in road vehicles	Guides the choice of a future 12 V adapter
ISO 7637-2 [67]	Transients on vehicle supply lines	Adapter must tolerate cranking dips and load dumps
UAE Personal Data Protection Law (PDPL) [68]	Processing of personal data in the UAE	Consent, data minimisation, no video upload
General Data Protection Regulation (GDPR) [69]	Data protection reference	Right to withdraw and delete dataset records
OpenStreetMap Overpass [70]	Points of interest query	Nearest rest stop or fuel station (Desired)

Chapter 4 Proposed Solution

4.1 What the System Must Do and Its Limits

4.1.1 System functions in plain words

A small camera watches the driver for four behaviours: yawning, covering the mouth, drinking and using a phone. The system ignores single odd frames and waits until a behaviour is held long enough to be real, then warns the driver with a light and a short voice message stating what it saw, escalating if the behaviour continues. Mirror checks remain silent. Every confirmed event is logged on the device, a display shows the current status and runs the calibration, and after the trip only event records reach a web dashboard of trends and tips, ensuring that no video ever leaves the device. In SDP 2, the system is demonstrated on a desk rig in a room, while the in-car placement remains the target for a vehicle version.

4.1.2 List of specifications

4.1.2.1 Functional specifications

Every Required and Desired feature of Section 1.4 is a numbered, testable specification (FS-01 to FS-26 in Tables 1.1 and 1.2), each mapped to a test in Section 5.4. Future features are discussed in Section 7.3.2.

4.1.2.2 Performance specifications

Table 4.1 sets a numeric target for the running system, along with how each target is measured.

4.1.3 List of constraints

Table 4.2 lists the constraints and assumptions that shape the design, along with the specifications each one affects. The system must assist the driver without ever replacing their attention, and a vehicle cabin is harsh on electronics, with power cuts, voltage dips and UAE heat. Throughout the

Table 4.1: Performance metric targets

Metric	Target	How measured
Frames per second on the Pi 5	25–40+ FPS	Frame counter averaged over a 30 min live run of the full pipeline after warm-up
End-to-end latency	$\leq$ 100 ms	Timestamps at frame capture and at the alert command, reporting the mean and 95th percentile
Per-class recall	0.90–0.95+	Confusion matrix on the subject-disjoint test set at the selected cutoffs
False alarms per hour	≤ 3 (target), < 10 (minimum)	Confirmed alerts on neutral and hard-negative test clips divided by their duration in hours
Camera-blocked detection	3 s	Lens covered or dazzled at a random moment, timed to the announcement over 10 trials
Eye-closure alert delay	1.5 s	Acted eye-closure clips on the demonstration rig, timed from the labelled closure start to the loud alert
PERCLOS window	60 s	Fixed in software and checked with synthetic EAR sequences
LED flash rate	$\leq$ 3 per second	Toggle timestamps from the log, checked with slow-motion video
Alert cool-down window	30 s	Two same-class events inside the window, where the second must not alert
Escalation timing	Tier 2 at 5 s, Tier 3 at 10 s	Persistent acted cue, with tier times read from the event log
Human response time to alert	$\leq$ 1.5 s	Acted trials on the demonstration rig, from alert start to the visible reaction in video

project, we follow the Institute of Electrical and Electronics Engineers (IEEE) [71] and Association for Computing Machinery (ACM) [72] codes of ethics and the UAE PDPL [68].

4.2 Alternative Solutions and How We Chose

4.2.1 Evaluation criteria and why they are ranked this way

We rank accuracy and false alarms first, since a system that misses events or keeps crying wolf ends up switched off [20, 4]. Table 4.3 gives each weight and how it is measured.

4.2.2 Alternative solutions for the whole system

4.2.2.1 Description of each alternative

Table 4.4 describes the five whole-system alternatives.

Table 4.2: Summary of design constraints

Constraint	Limit / requirement	Effect on the design	Related spec ID
Economic	5000 AED budget	Required parts first, and the request totals 2725 AED	FS-01 to FS-10
Safety	≤ 3 flashes/s, peripheral light, safe volume, clear view	Flash rate capped in software, night dimming, mount away from line of sight	FS-05, FS-14, FS-15
Security and privacy	No video out, Personal Identification Number (PIN) identity, authenticated TLS	On-device inference, event-only logs, device key	FS-08, FS-09, FS-19, FS-22
Reliability	Power cuts, heat, vibration, glare	SQLite WAL, active cooler, blind check	FS-17, FS-21
Power and size	5 V / 5 A, ≤ 25 W	Power budget with margin, compact vented case	Section 4.3.3.4
Usability	No interaction while driving, bilingual voice	Short messages, glanceable UI, dark mode	FS-16, FS-24
Ethical and social	Consent, fairness, no surveillance	Authors as subjects, public data for appearance variety, driver owns the data	FS-10, FS-19
Assumption: setup	Fixed mount, face in view, no sunglasses, enough cabin light, internet only after the trip	RGB camera only, SDP 2 testing and demonstration on a desk rig in a room	FS-02, FS-03
Assumption: scope	Four subjects, four classes, no head-pose model, not certified for vehicles	Visual distraction without an object is not detected	FS-10

Table 4.3: Evaluation criteria, weights and how each is measured

Criterion	Why it matters	Weight	How measured
Accuracy and false alarms	Missed events defeat the purpose, false alarms make drivers switch the system off	20%	Recall and false alarms per hour
Real-time speed on the Pi	A late alert is useless, slow frames miss short cues	18%	FPS and latency
Privacy	Video of the driver's face is personal data	15%	Whether video leaves the car
Cost	5000 AED budget, low unit cost helps adoption	14%	Price in AED
Power and heat	Continuous operation in UAE cabin heat	12%	Watts and temperature
Ease of installation	No special tools or wiring skills	8%	Parts and steps
Complexity	More parts mean more failure points	7%	Build effort estimate
Parts availability	Local purchase keeps the schedule	6%	Stock in the UAE

Table 4.4: Whole-system alternatives

ID	Alternative	How it works	Main weakness
A	On-device object detection	YOLO11n on the Pi 5 with the AI HAT+, alerts made in the car	Needs our own dataset, optimisation and cooling
B	Cloud-streamed vision	The car uploads video and a large model runs on a server	Needs a constant connection, adds delay, face video leaves the car
C	Landmark rules	EAR, mouth opening and head angle with fixed thresholds	Cannot see a phone or a bottle
D	Vehicle signals or wearables	Steering, lane position or heart rate	Indirect, cannot tell what the driver is doing
E	Smartphone app	The phone's camera and processor	Overheats on a sunny dashboard, limited background camera use

4.2.2.2 Comparison and selected approach

Figure 4.1 scores each alternative from 1 to 5. Alternative A achieves the highest total (3.96), narrowly ahead of C (3.93). However, C cannot detect phone use, drinking or hand-over-mouth, failing the core requirement FS-03 regardless of its score. The weak points of A are accepted, as each can be handled within SDP 2.

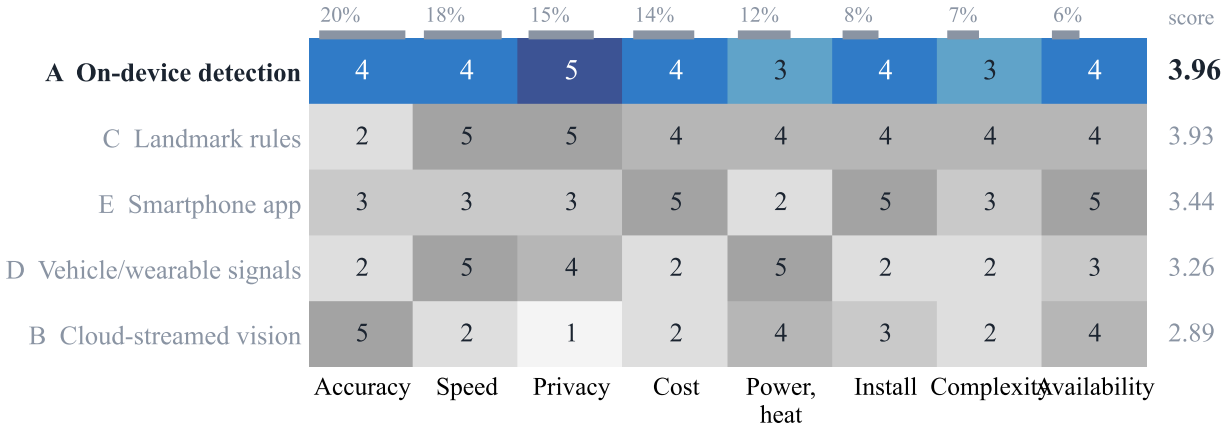


Figure 4.1: Decision matrix for the whole-system alternatives (scores 1–5, weights above each criterion), sorted by total with the selected design in colour

4.2.3 Alternatives for each major part

4.2.3.1 Processing platform

On the Pi 5 CPU alone, YOLO11n reaches only 6.79 FPS [55], so we add the AI HAT+ to move detection onto an NPU. Table 4.5 compares the options.

Table 4.5: Processing platform options

Option	Price (AED)	Power (W)	Expected FPS	UAE availability	Verdict
Raspberry Pi 5 (CPU only)	845	5–12	6.8 at 640 [55]	Good	Fallback only
Pi 5 with AI HAT+	845 + 391	+2–3	$\geq$ 30 (to be measured)	Good	**Selected**
Jetson Orin Nano	About 900–1100	7–25	$\geq$ 30	Limited	Too costly, more heat
Raspberry Pi 4	About 300	4–7	Below 5	Good	Too slow

4.2.3.2 Camera

A USB RGB webcam is plug-and-play and matches the public RGB data we pre-train on. An IR camera performs better at night and remains a Future item (Table 4.6).

Table 4.6: Camera options

Option	Resolution / FPS	Night performance	Cost (AED)	Verdict
USB UVC RGB (C922)	1080p at 30	Poor without cabin light	255	**Selected**
IR / NIR camera	720p–1080p at 30	Good with illuminators	150–400	Future
Pi camera (CSI)	Up to 12 MP	Good with NoIR variant	About 120–150	Short ribbon, awkward to mount
Dashcam	1080p+	Varies	200+	Closed firmware, no live stream

4.2.3.3 Detection model

YOLO11n offers the best balance of size and accuracy among models with a mature path to the Hailo NPU, with YOLO26n kept as a backup should its NPU export mature (Table 4.7).

Table 4.7: Detection model candidates (COCO, 640 px)

Model	Params (M)	GFLOPs	$\mathbf{mAP}_{50-95}$	Pi 5 CPU FPS	Licence
YOLO11n [52]	2.6	6.5	39.5	6.8 [55]	AGPL-3.0
YOLOv8n	3.2	8.7	37.3	SDP 2	AGPL-3.0
YOLO11s	9.4	21.6	47.0	SDP 2	AGPL-3.0
YOLOv10n [73]	2.3	6.7	38.5	SDP 2	AGPL-3.0
YOLO26n [74]	2.4	5.5	40.9	7.8 [55]	AGPL-3.0

4.2.3.4 Remaining components

Table 4.8 lists the remaining parts. A light bar paired with a voice message is preferred over a buzzer, which cannot explain itself, and over a screen, which pulls the eyes off the road [30, 32].

The PXN wheel serves as the steering wheel of the desk demonstration rig and prototypes a Future vibration alert.

Table 4.8: Remaining components, alternatives and electrical specifications

Part	Selected (AED)	Alternatives	Why selected	Key electrical spec
Audio alert	Creative Pebble V3 (177)	Buzzer, 3.5 mm speaker with amplifier	Clear voice messages, no amplifier board, USB powered	5 V USB, USB audio class
Visual alert	RGB light bar (70)	Single LED, strobe beacon	Visible in peripheral vision, one colour per class	5 V, about 0.3 A, switched by Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET)
Driver display	Hosyond 7-inch touch (245)	Phone over Wi-Fi, 20×4 character Liquid Crystal Display (LCD)	Room for status and calibration screens, no extra radio	5 V, about 0.5 A, High-Definition Multimedia Interface (HDMI) and USB touch
Power supply	Official 27 W PD (112)	5 V / 3 A supply, power bank	Only a 5 A supply raises the USB limit to 1.6 A	5.1 V, 5 A, USB-C PD
Storage	SanDisk Extreme Pro 128 GB (136)	USB SSD, basic microSD	Fast and durable, low write load from events only	A2, U3 microSD
Cooling and case	Active Cooler (64), metal case (86)	Passive heatsink, plastic case	Keeps the Pi below throttling under NPU load	Fan powered and controlled by the Pi
USB hub	UGREEN 3.0 hub (28)	Pi ports only	Four USB devices on the Pi's ports	Bus powered, shares the 1.6 A
Light switch	Logic-level MOSFET (lab stock)	Relay module, BJT	Silent, fast Pulse-Width Modulation (PWM), fully on at 3.3 V gate drive	Gate threshold 1–2 V
Server and maps	FastAPI, PostgreSQL and React on a free tier	Paid map Application Programming Interface (API), managed backend	Free and open source, and Overpass needs no key [70]	HTTPS with TLS 1.3

4.3 The Selected Design: Technical Approach

4.3.1 System overview: the big picture

4.3.1.1 Overall block diagram

Figure 4.2 presents the whole system. Before deployment, the model is trained, hardened with hard negatives and compiled for the NPU. During a session, frames flow from the camera through the detector and the EMA into the decision block, the only place where scores turn into alerts. After the session, only event records leave the device. The driver display acts as both an output and an input, since calibration writes the cutoffs that the decision block uses.

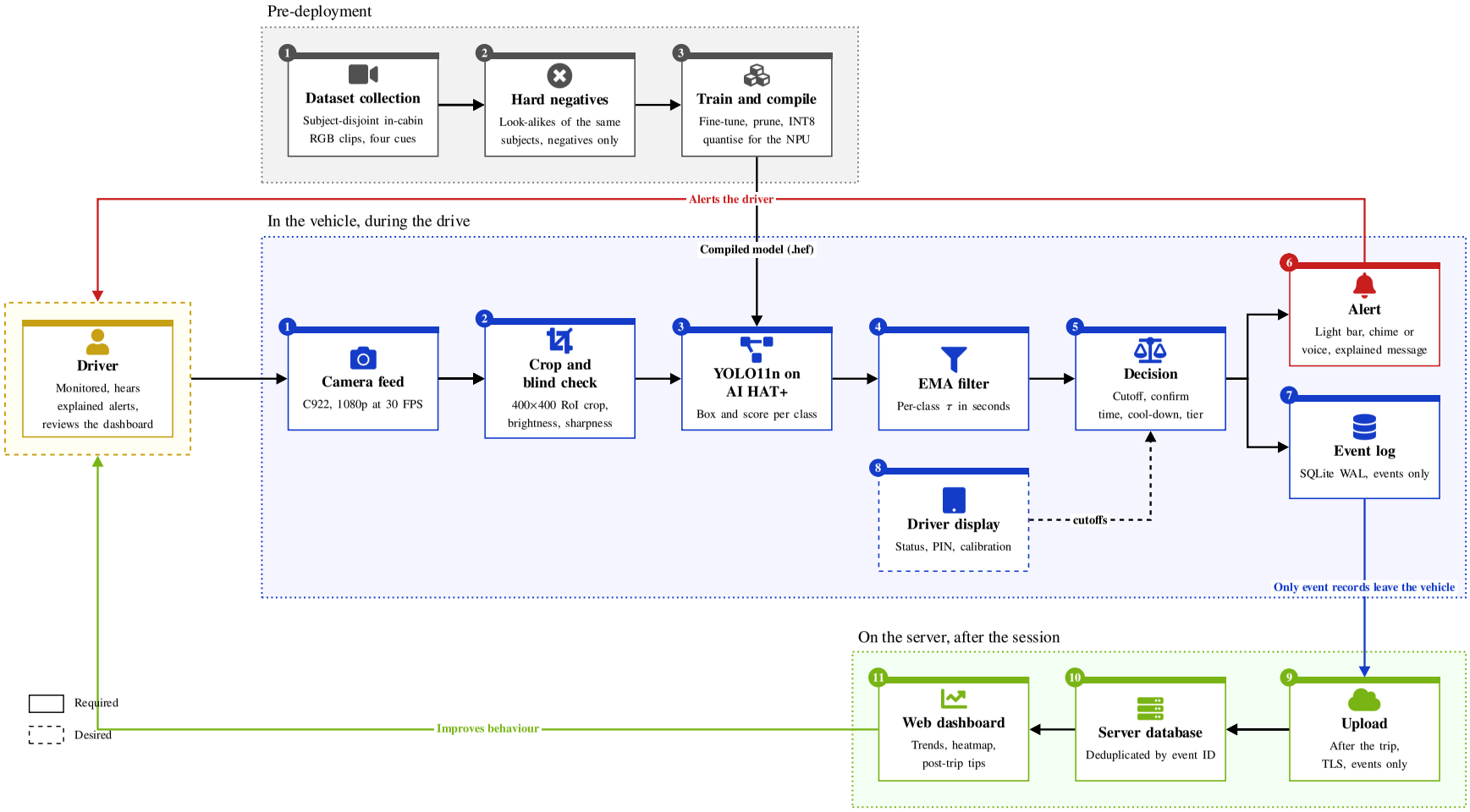


Figure 4.2: System block diagram, where only event records cross from the vehicle to the server

4.3.2 How the system operates step by step

At power-on, the Pi checks its devices, the driver enters a PIN, and a new driver completes the 20 s calibration of Section 4.3.4.6 (Figure 4.3). Each frame is then checked, cropped, scored and smoothed, and a cue is confirmed only once it stays above the cutoff for its confirmation time (Figure 4.4). The system fails safe: a blocked view is announced, missing eyes are skipped rather than counted as open, and a power cut never corrupts the log. After the session, unsent events are uploaded over TLS whenever Wi-Fi is available.

4.3.3 Hardware design in detail

4.3.3.1 Hardware architecture and component selection

The Raspberry Pi 5 (8 GB) boots from a 128 GB microSD card. The Active Cooler sits on the SoC, while the AI HAT+ stacks above it on brass standoffs and connects to the Pi through a short PCIe ribbon, with a vented metal case holding the whole stack. The camera, speakers, touch screen and wheel connect through a USB 3.0 hub, the display additionally uses micro-HDMI, and the light bar is switched from General-Purpose Input/Output (GPIO)18. Figure 4.5 shows the complete bench, Figure 4.6 shows the stack layer by layer, and Table 5.9 lists every part.

4.3.3.2 Demonstration rig for SDP 2

The SDP 2 demonstration takes place in a room rather than in a car. The driver sits at a desk behind the PXN wheel, the C922 stands above the wheel at dashboard height facing the driver's face and hands, the light bar runs along the desk edge within peripheral vision, and the speakers, the 7-inch display and the edge unit sit on the desk, as arranged in Figure 4.5. This rig reproduces the geometry of the in-car mount while keeping power on mains, the temperature controlled and every test repeatable. Since our dataset is recorded in parked cars, the room is a new setting for the model, which we address through the per-driver calibration at the start of each demonstration and the brightness and hue augmentation of Table 4.10.

4.3.3.3 Circuit and wiring diagram with all connections

A GPIO pin cannot supply the current the light bar needs, so a logic-level N-channel MOSFET switches the ground side of the bar. GPIO18 (pin 12) drives the gate through a 220 Ω resistor, while a 10 kΩ pull-down keeps the bar off as the Pi boots. The bar takes 5 V from pin 2, and the source joins the common ground at pin 6. All other parts use standard USB, micro-HDMI, PCIe and USB-C connectors. Figure 4.7 shows the wiring alongside its schematic.

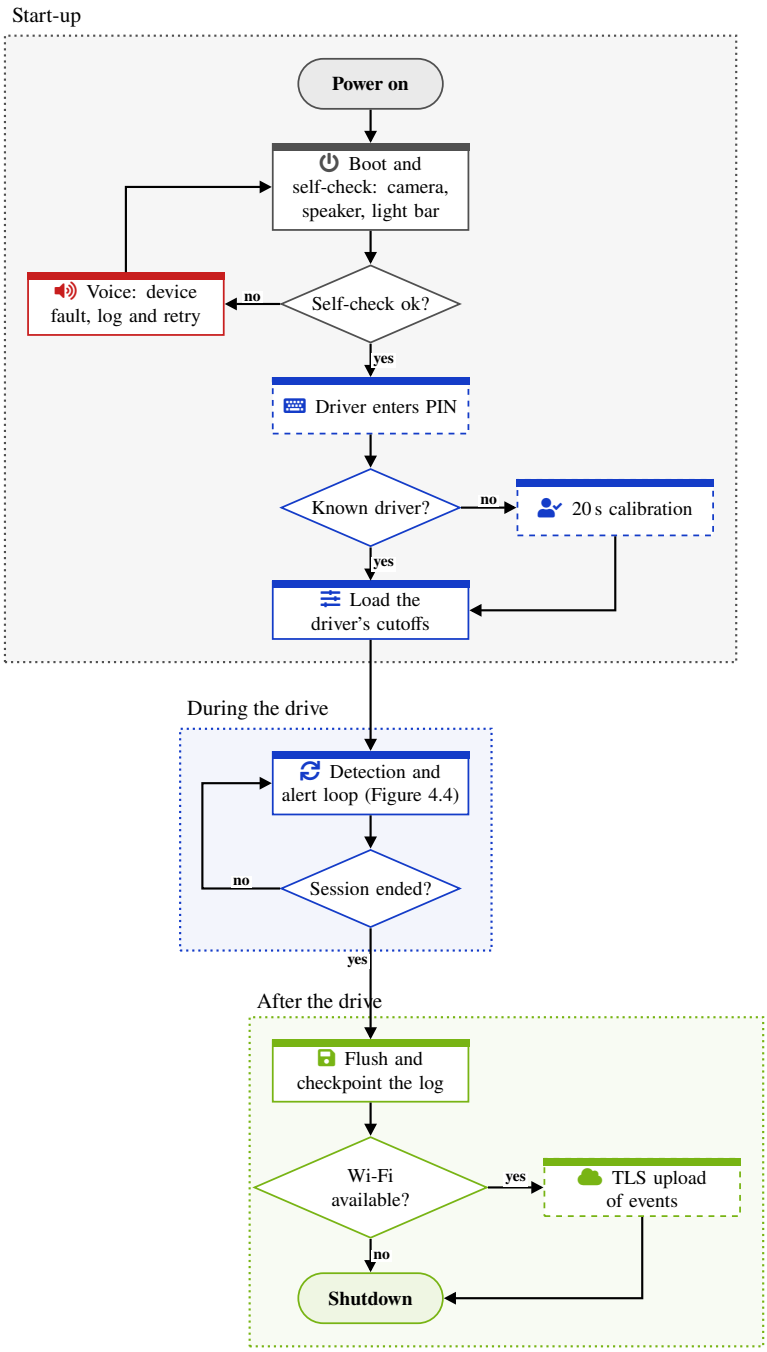


Figure 4.3: Top-level system flowchart (dashed: Desired)

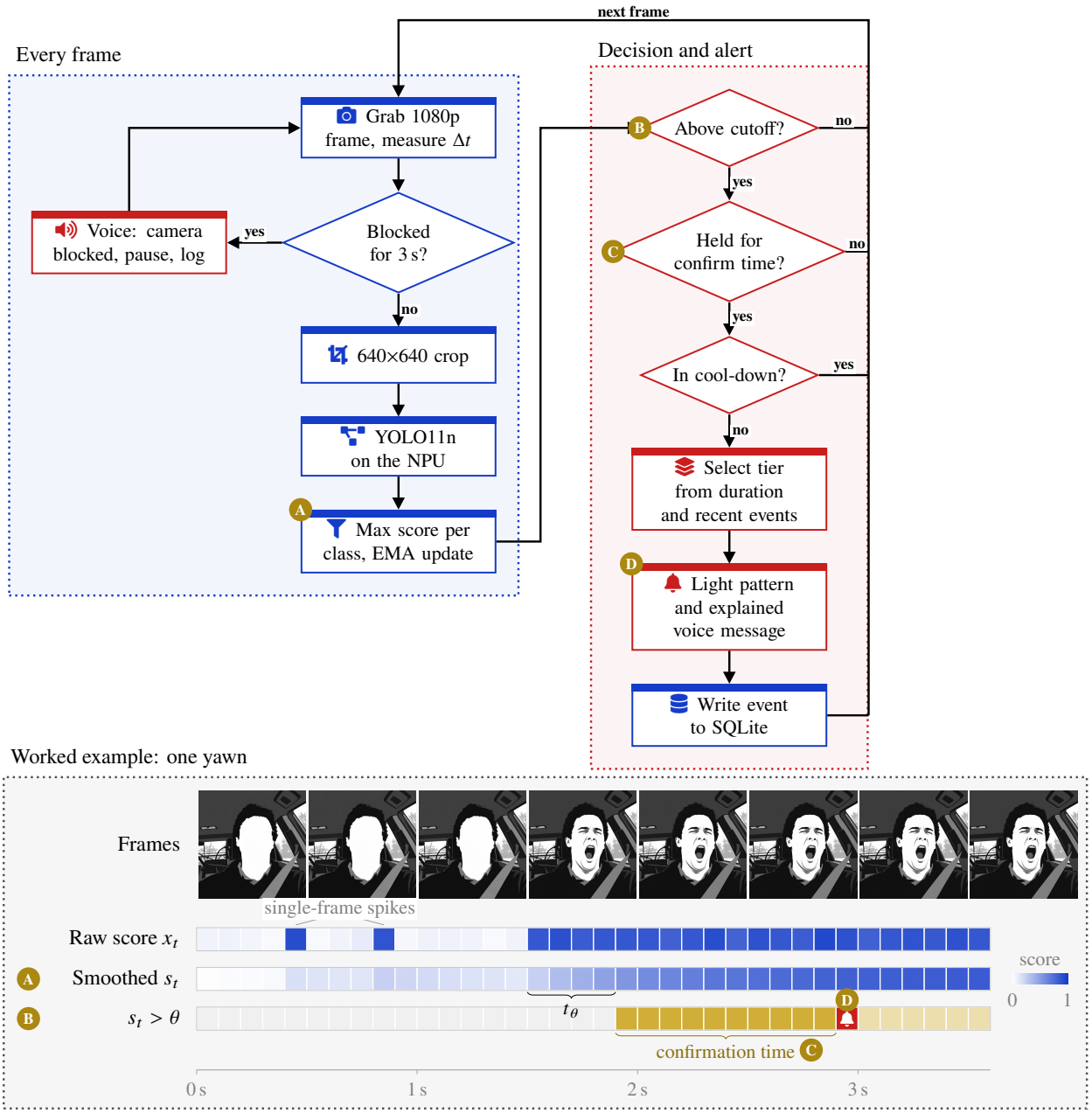


Figure 4.4: Detection-and-alert loop with a worked example for one yawn at the Table 4.11 settings, drawn at 10 FPS with illustrative scores. Letters tie each step to the row that shows it

4.3.3.4 Power chain and power budget

All loads share the 5.1 V, 5 A rail of the official 27 W supply, with the USB ports sharing 1.6 A [54]. Table 4.9 estimates a typical draw of 3.25 A (16.3 W), leaving about 9 W of margin. In the worst case, with every load at its peak, the draw slightly exceeds both 5 A and the USB limit, so we cap speaker volume and light brightness in software, powering the hub separately should bench tests confirm the issue. The demonstration rig runs from mains, while an in-car version would use

Figure 4.5: SDP 2 hardware bench with all purchased parts and a magnifier showing the stack inside the case (Blender render)

a 12 V PD adapter rated for ISO 7637-2 transients [67].

4.3.3.5 Target in-car placement

In a vehicle, which remains the target of the product beyond SDP 2, the camera sits on the dashboard above the steering column, capturing the face and both hands of the driver. The light bar runs along the dashboard edge within peripheral vision, the display sits by the centre vents, within reach but outside the line of sight, and the edge unit rests on the centre console, where it is shaded and ventilated. Figure 4.8 shows this placement next to a real view from one of our pilot sessions, while Section 5.3.2.2 explains why the camera does not go below the wheel.

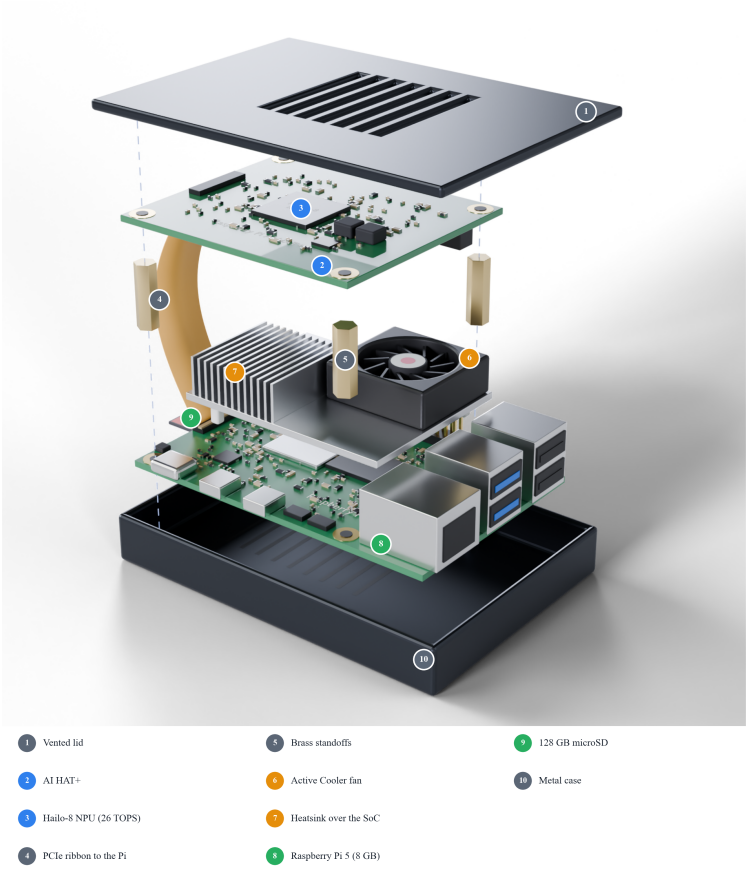


Figure 4.6: Edge unit, exploded from the case to the lid (Blender render)

Table 4.9: Estimated power budget (5 V rail), to be measured in SDP 2

Component	**Voltage**	**Typical A**	**Peak A**	**Watts (typ.)**
Raspberry Pi 5 with active cooler	5 V	1.20	2.00	6.0
AI HAT+ (Hailo NPU)	5 V	0.50	0.60	2.5
Logitech C922 (USB)	5 V	0.25	0.50	1.3
Creative Pebble V3 (USB)	5 V	0.40	1.00	2.0
7-inch display and touch	5 V	0.50	0.80	2.5
RGB light bar via MOSFET	5 V	0.30	0.60	1.5
USB hub and PXN wheel	5 V	0.10	0.30	0.5
Total	5 V	**3.25**	**5.80**	**16.3**

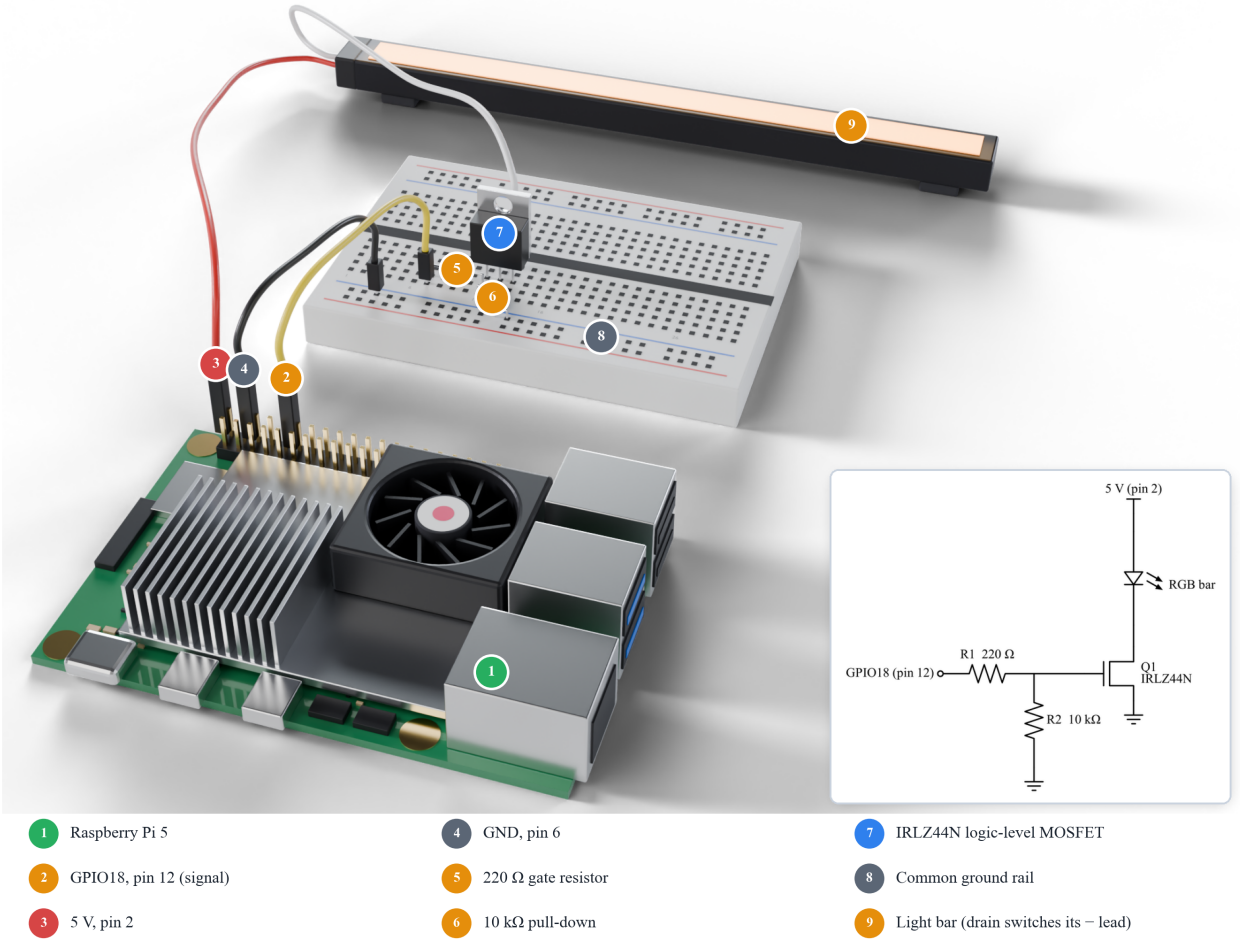


Figure 4.7: Light-bar driver on the breadboard, with its schematic (Blender render)

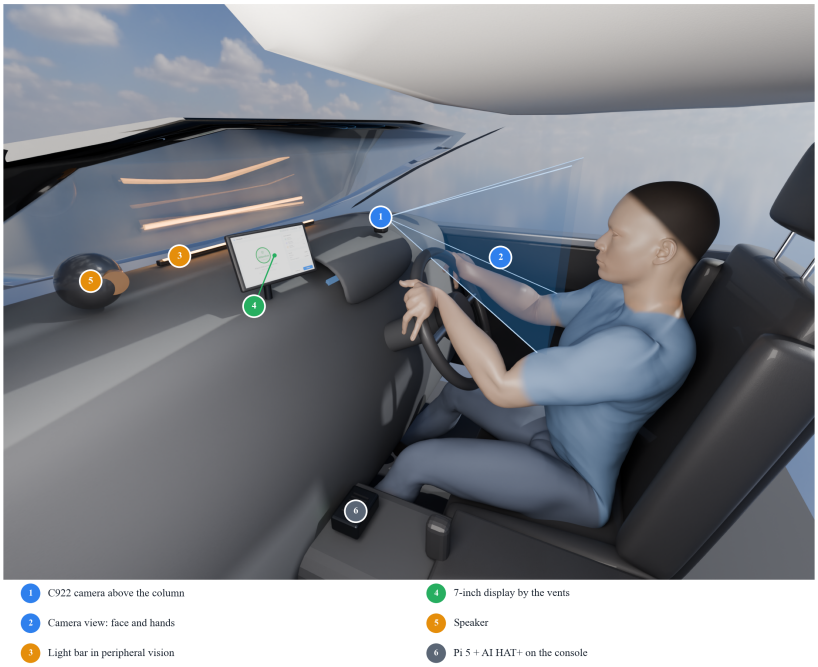
4.3.4 Software design in detail

4.3.4.1 Software architecture and module hierarchy

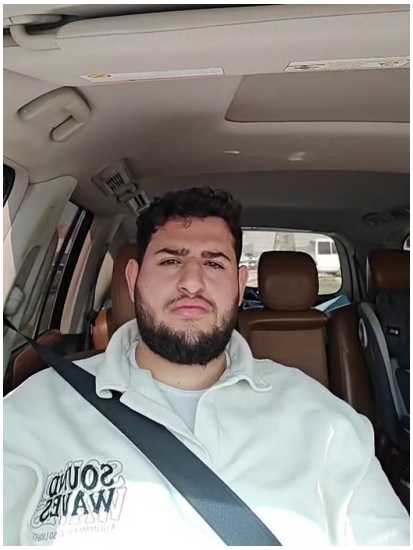
The software consists of small Python modules connected by queues (Figure 4.9). A capture thread passes on only the latest frame, so a slow stage never builds a backlog. The detector runs in its own thread on the NPU, the filter, decision logic and alerts share the main loop, and the logger, uploader, display server and eye module run in their own threads. Each Desired feature plugs into this loop without changing it, allowing features to be added or removed one at a time.

4.3.4.2 Training and optimising the detector

YOLO11n is fine-tuned from its COCO weights on the subject-disjoint training split at an image size of 640, with augmentation for lighting, small rotations and horizontal flips. The validation



(a) Target in-car placement (Blender render, driver from the Blender Studio CC0 base meshes)



(b) Real view from a dash mount (pilot session C3)

Figure 4.8: Target in-car placement and what the camera sees

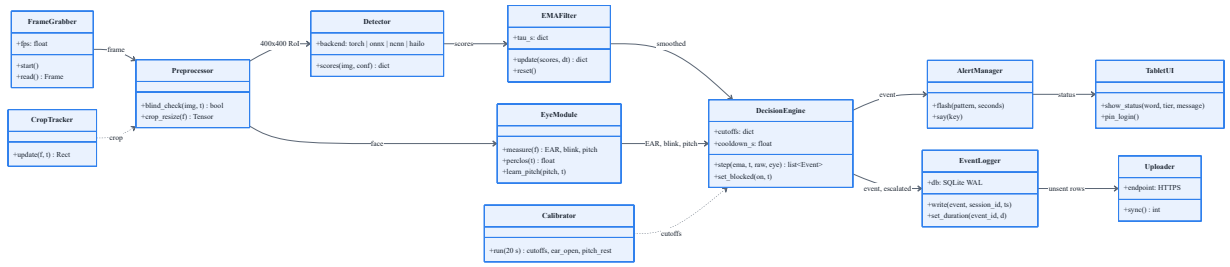


Figure 4.9: UML class diagram of the main software modules

split is used for early stopping and for choosing cutoffs, while the test split is touched only once, at the very end. Table 4.10 lists the settings.

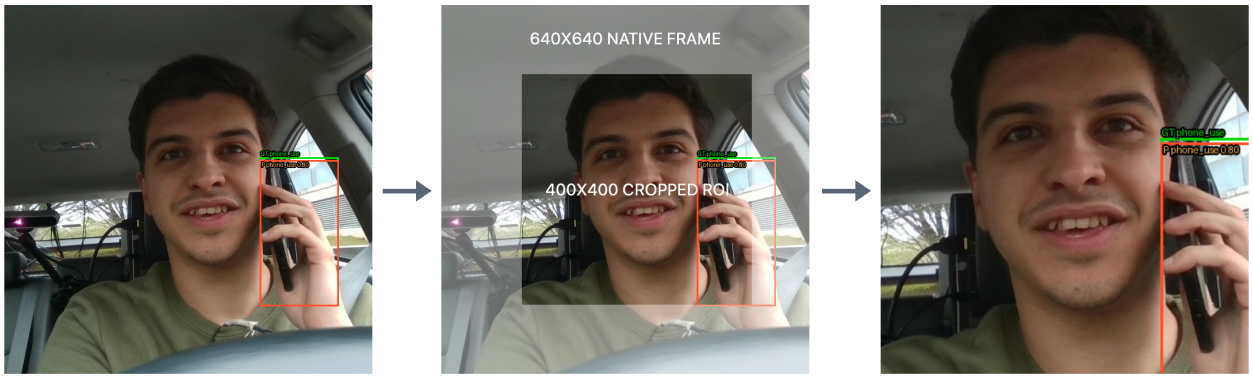
Compression follows a fixed order: fine-tune, prune in 10% steps, fine-tune again, quantize to INT8 with the Hailo Dataflow Compiler, and finally benchmark on the Pi. A variant is acceptable when recall drops by no more than 2 points and mAP_{50} by no more than 3 points against the FP32 model, and the fastest acceptable variant is deployed (Table 4.10).

Table 4.10: Training settings and compression configurations

Stage	Setting	Value
Fine-tuning	Initial weights, image size	COCO YOLO11n, 640
Fine-tuning	Augmentation	Brightness and contrast, hue, ±10° rotation, horizontal flip, mosaic
Fine-tuning	Epochs, early stopping	Up to 150, patience 30 on validation mAP_{50}
Pruning	Structured, by channel L1 norm	10%, 20%, 30% then fine-tune
Quantization	Methods compared	Hailo INT8 (PTQ), ONNX FP16 and INT8, NCNN INT8
Selection	Acceptable	Recall drop ≤ 2 points, mAP_{50} drop ≤ 3 points

4.3.4.3 Region-of-interest crop

Each frame is first cut to a 640×640 square, from which a fixed 400×400 RoI at $x \in [140, 540]$, $y \in [120, 520]$ is extracted and resized to the model input (Figure 4.10). Since the driver occupies a predictable region of an in-cabin view, the RoI keeps the face and hands, discards 61% of the pixels, and enlarges small cues: at a 320-pixel input, the median yawning box grows from 35 to 55 pixels [75]. Detections are then mapped back to the full frame. The cost falls on the edges of the frame, as drivers tend to hold a phone toward the door, so 9.3% of test phone-use boxes in DMD leave the window [75]. Crop tracking (Table 4.12) moves the window with the face to reduce this loss.



1. Native 640×640 frame 2. Fixed 400×400 RoI 3. RoI resized for the model

Figure 4.10: Static RoI crop on a DMD frame: the phone box (ground truth in green, prediction in red) stays inside the window and gets larger (from [75])

4.3.4.4 Temporal filtering with EMA

Each class has its own τ: 0.6 s for yawning and hand-over-mouth, which are short and distinct, and 1.0 s for drinking and phone use, which last longer and have more look-alikes. The filter restarts from zero after boot and after a camera-blocked period, ensuring old evidence never carries over.

Since α is computed from the measured Δt in Equation 3.1, the filter is tested by replaying the same clip at 10, 15 and 30 FPS and checking that confirmation times agree within one frame.

4.3.4.5 Alert decision logic: class-specific alerts, cool-down, escalation and suppression

Each class has a default cutoff and, after calibration, a per-driver cutoff kept between the default and 0.85. A cue is confirmed once its smoothed score stays above the cutoff for the class confirmation time. When the cue ends, the class is muted for a 30 s cool-down, and any cue inside the cool-down is logged as tier 0 without playing an alert. Counting from the end of the cue allows Tiers 2 and 3 to fire while a cue persists, whereas the eye-closure alarm has no cool-down, being the most dangerous case. Each class has its own light pattern and sound, always below 3 flashes per second. We designed three alarm sounds in Matrix Laboratory (MATLAB): a gentle 1.5 s chime for distractions, a 3 s 350 Hertz (Hz) buzz for yawning, and a 10 s emergency alarm reserved for risky driving. Table 4.11 lists the per-class settings.

Alerts escalate in three tiers while the risk persists (Figure 4.11). Tier 1 is the class chime with a short light pulse. Tier 2 adds an explained voice prompt, after 5 s or after three cues within 5 minutes. Tier 3 is the emergency alarm with the full light pattern, after 10 s, or immediately when the eyes stay closed for 1.5 s (3 s with the head down). The PERCLOS warning acts as a gentle Tier 2 prompt. Every voice message names what was detected and for how long or how often, in both English and Arabic. Since a single MOSFET can flash the bar but not change its colour, the colours in Table 4.11 appear on the display instead.

Mirror checks never trigger an alert, as head orientation is not one of our classes. Suppression while reversing, at red lights or in heavy traffic requires vehicle signals, and therefore remains a Future item.

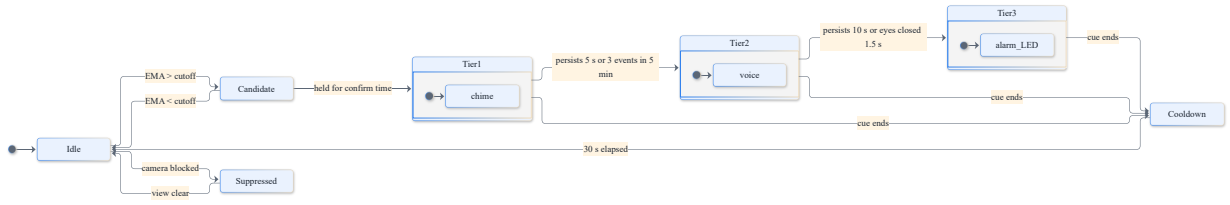


Figure 4.11: Alert state machine with the three tiers, cool-down and suppression

4.3.4.6 New-driver calibration algorithm

During a 20 s window, guided by prompts on the display, the driver sits normally while the detector runs at a confidence of 0.05. Since the driver performs no cue in this window, every detection

Table 4.11: Per-class alert settings

Class	Default cutoff	τ (s)	Confirm (s)	Light pattern	Voice message (Tier 2)
Yawning	0.50	0.6	1.0	Amber, 1 Hz pulse	"Yawning detected 3 times in 5 minutes. Consider a short break."
Hand-over-mouth	0.50	0.6	1.0	Amber, 1 Hz pulse	"Hand over mouth detected for 5 seconds. You seem tired. Consider a short break."
Drinking	0.55	1.0	1.5	Teal, single pulse	"Drinking detected. Keep your eyes on the road."
Phone use	0.55	1.0	2.0	Purple, 2 Hz	"Phone use detected for 5 seconds. Please put the phone down."
Eyes closed	0.50 × EAR_{open} and blink score > 0.50	–	1.5 (3.0 head down)	Red, 3 Hz	"Eyes closed for 1.5 seconds. Pull over safely." (Tier 3)

within it is treated as a false alarm. For each class c, the cutoff becomes

$$\theta_c = \min\Big(\max\Big(P_{99}(\text{scores}_c) + 0.05,\ \theta_c^{\text{default}}\Big),\ 0.85\Big). \tag{4.1}$$

We define "many detections" as more than 10% of the 600 baseline frames scoring above the default cutoff in any class, indicating that the driver moved or performed a cue, in which case the defaults are kept and the driver is asked to redo the baseline. The same window also stores the driver's open-eye EAR and rest head pitch. Figure 4.12 shows two example drivers and the flow.

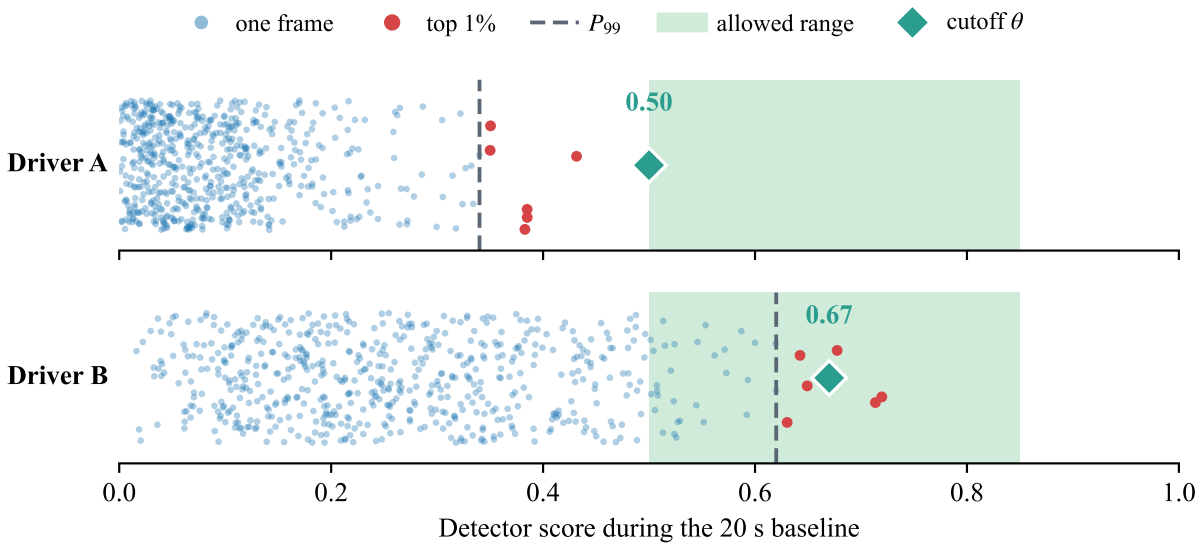
4.3.4.7 Desired driver-facing features

The remaining Desired features each add one small step to the loop of Figure 4.4 (Table 4.12). Two rules apply to all of them: missing data is never treated as good news, so frames without visible eyes are skipped, and the driver never has to touch the display while moving. Figure 4.13 shows the eye-closure logic and Figure 4.14 the display screens.

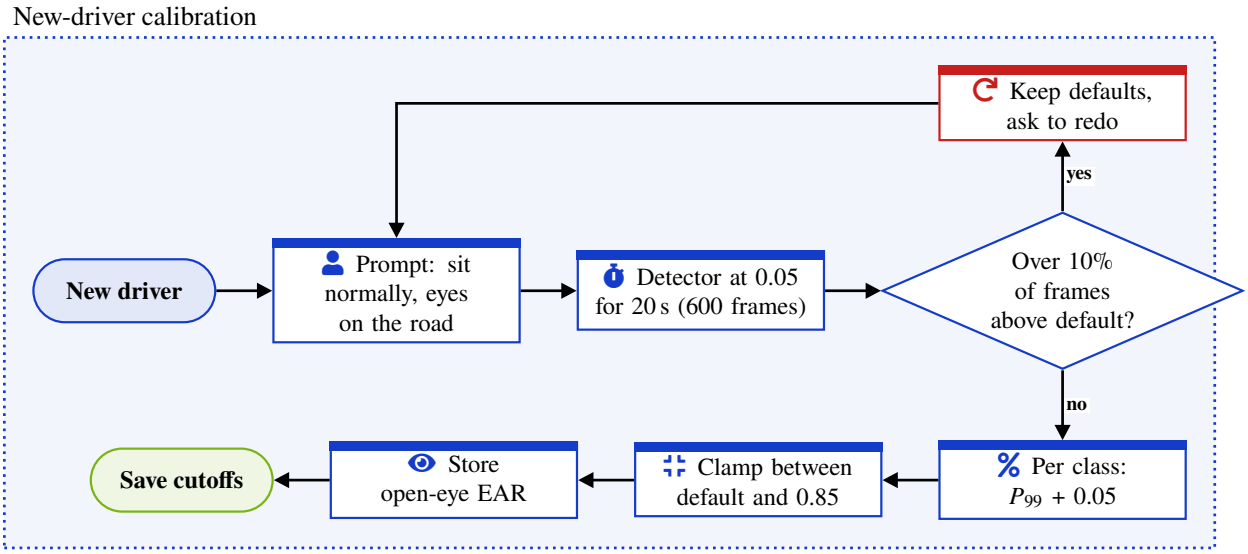
4.3.5 Logging, server and dashboard

4.3.5.1 Power-cut-safe on-device logging

Events are stored in SQLite with WAL mode and `synchronous=NORMAL`, committing each event in a single short transaction, so a power cut cannot corrupt the database, although it may lose the last commits. The schema holds driver, session, event, calibration and tip tables (Figure 4.15), where an event row stores the time, class, confidence, duration, tier and message, along with `uploaded` and



(a) Two example drivers: the cutoff is P_{99} plus 0.05, kept inside the allowed range (illustrative)



(b) Calibration flow

Figure 4.12: New-driver calibration: the cutoff is the 99th percentile plus 0.05, kept between the default and 0.85

`escalated` flags. No frames are stored by default. On every boot, `PRAGMA integrity_check` runs before a new session starts, and a failed check moves the old file aside and starts a fresh one.

4.3.5.2 Session upload and server update

After the session, the uploader selects unsent events and posts them as JavaScript Object Notation (JSON) to the server over HTTPS with TLS 1.3, authenticated by a per-device key in a request

Table 4.12: Desired driver-facing features

Feature	How it works	Key values	Spec
Camera-blind check	Mean grey brightness and variance of the Laplacian [76] per frame. When unclear for 3 s, it says "camera blocked", pauses alerts and logs the period, which feeds the % of drive time monitored	Brightness < 25 or > 230, variance < 20, tuned on covered-lens and torch tests	FS-17
Crop tracking	OpenCV YuNet face detector on a 480×270 copy, and the crop centre follows the face through its own EMA and is clamped at the frame borders. It is kept only if it beats the fixed crop	About 1 Hz, τ = 1 s, 640×640 window	FS-18
Eye closure	MediaPipe Face Mesh [77] landmarks give the EAR of both eyes, and an eye is closed below 50% of the driver's open-eye value with the model's blink score above 0.50. Head pitch more than 8° below the rest value means the driver may only be looking down	1.5 s closure → Tier 3, 3 s with the head down, PERCLOS > 15% over 60 s → gentle prompt	FS-20
Driver display	Full-screen web page served by the Pi: one status (Attentive, Distracted or Drowsy), recent events and trip time, a dark theme after sunset and touch only for PIN, calibration and settings	7-inch, 1024×600	FS-24
Pre-trip check	Two questions before driving (hours slept, how tired), where a tired answer leads to a suggestion to rest	Two taps	FS-25
Rest-stop suggestion	Nearest rest stop, fuel station or café from the paired phone's location (browser geolocation, no GPS module) through OpenStreetMap [70], with the location used once and not stored	Alerts in the last 30 minutes above a threshold	FS-26

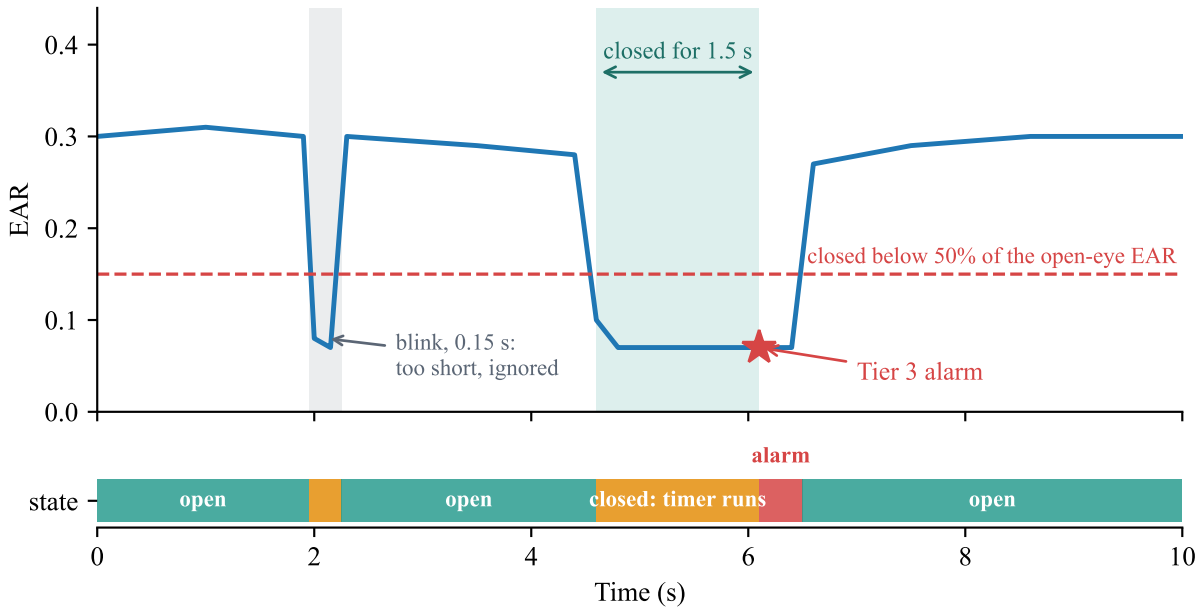
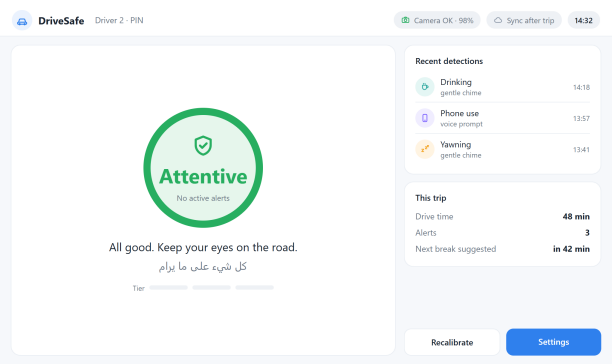
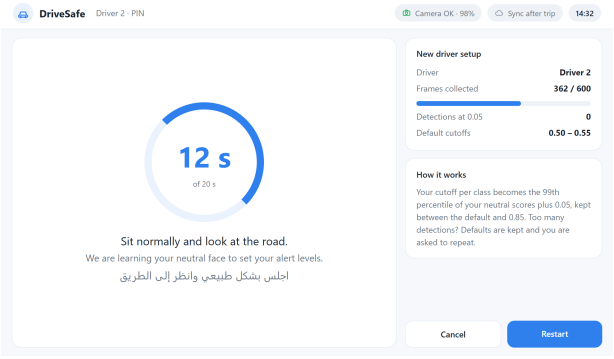


Figure 4.13: Eye-closure rule: a short blink is ignored, eyes closed for 1.5 s start the Tier 3 alarm (illustrative signal)



(a) Attentive (day)

(b) Drowsy, Tier 2 (night)



(c) New-driver calibration

Figure 4.14: Driver-facing display screens (UI mock-ups)

header. Each event carries a random identifier created on the device, allowing the server to ignore duplicates when an upload is retried, and the device marks rows as uploaded only after the server confirms. Failed attempts back off for 30, 60 and 120 s before waiting for the next session. No frames, embeddings or images are ever sent. Figure 4.16 shows the sequence.

4.3.5.3 Web dashboard and post-trip tips

The dashboard presents the driver's total drive time, distraction and drowsiness counts, the percentage of drive time monitored, a trend line, a breakdown by class, an activity heatmap by weekday and hour, and a timeline of the last trip. Access is limited to the driver's own account, protected by its own password, since PINs never leave the device. Post-trip tips are rules over recurring patterns, for example "yawning rose after 80 minutes on 4 of 7 trips: take a break around 90 minutes", or "phone use happened on 5 of your last 7 trips: set navigation before leaving". Figure 4.17 shows the design.

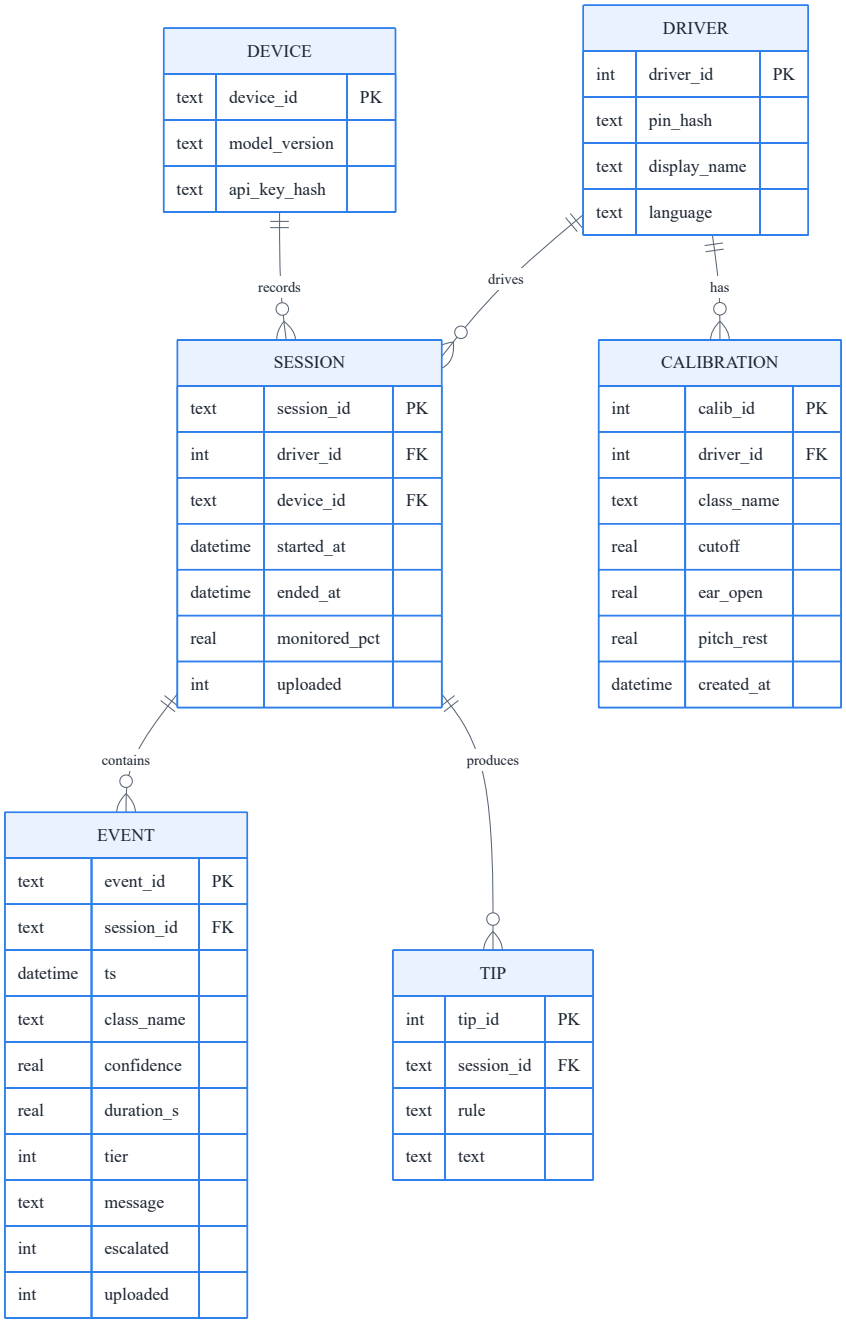


Figure 4.15: Entity-relationship diagram of the event database (same schema on the device and the server)

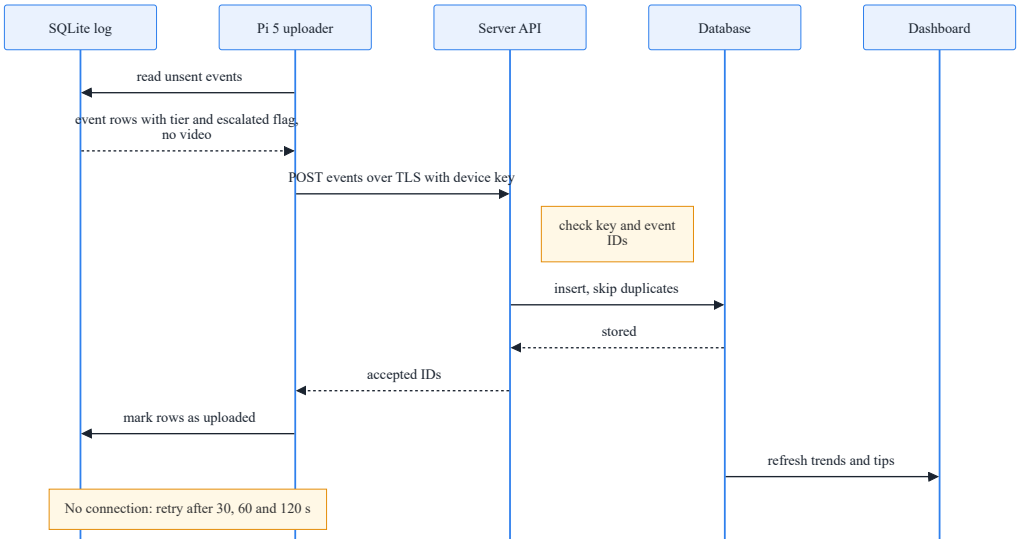


Figure 4.16: Upload sequence from the Pi to the server and dashboard

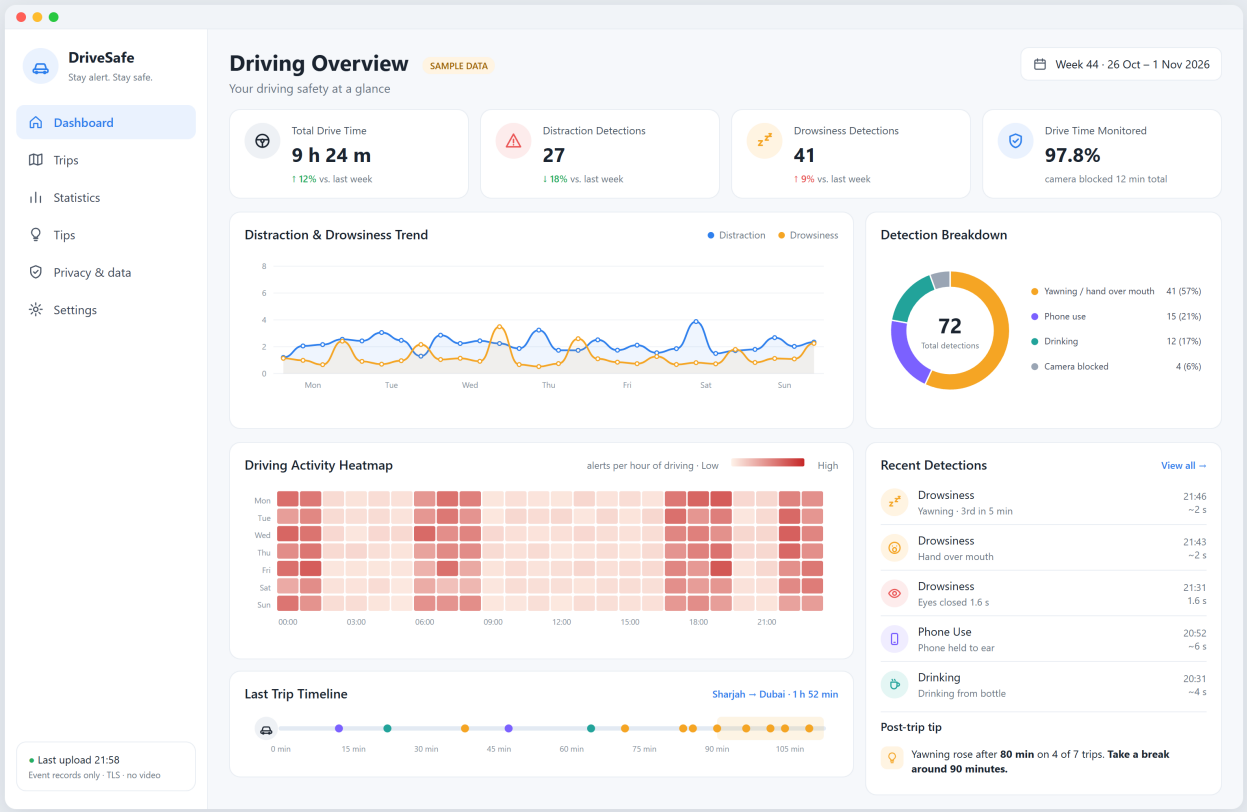


Figure 4.17: Web dashboard design (sample data)

4.3.6 Calculations, simulations and preliminary results

4.3.6.1 Calculations that justify the design values

Table 4.13 gives the calculations behind the main design values.

Table 4.13: Calculations that justify the design values

Topic	Calculation	Design effect
EMA	With $\tau = 0.6$ s, $\alpha = 0.054$ at 30 FPS and 0.105 at 15 FPS. One spike of $x = 1$ gives only 0.054. Reaching 0.5 from zero takes 13 frames, so $t_\theta = -0.6 \ln 0.5 = 0.42$ s (Equation 3.2)	α must come from Δt, and single spikes never alert
Compute	6.5 GFLOPs × 30 FPS = 195 GFLOP/s, while the Pi CPU gives about 44 GFLOP/s (6.79 FPS [55]). At 320 px the work drops to 1.6 GFLOPs, close to 27 FPS	The AI HAT+ is Required, with a 320 px CPU fallback
Storage	About 120 bytes per event, so 100 events per day add 4.4 MB per year, and an upload of 100 events is about 20 kB	The 128 GB card and Wi-Fi are far more than enough
Power	5 V × 3.25 A = 16.3 W typical on a 25.5 W supply (Table 4.9)	About 9 W of margin

Latency is measured from frame capture to the alert command, excluding the deliberate confirmation time. Table 4.14 sets a 60 ms target against the 100 ms limit.

Table 4.14: Latency budget per stage

Capture, crop	Detection (NPU)	NMS	EMA and decision	Alert output	End-to-end
8 ms	20 ms	4 ms	2 ms	26 ms	**60 ms**

4.3.6.2 Preliminary experiments

Before the SDP 2 hardware arrives, we conducted four studies on public data using a workstation with an RTX 4060 GPU, each released as an open repository [78, 50, 75, 79]. Table 4.15 summarises what each study measured and the design decision it informed.

Detector choice. On DMD with a 14-driver subject-disjoint split, YOLO11n achieved the highest mAP_{50} (0.918) and speed (68 FPS) of three nano detectors [78]. Even so, its F1 ranged from 0.62 to 0.95 between test drivers. The false positive in Figure 4.18, a tissue held over the mouth, is precisely the kind of look-alike our hard-negative round targets.

Figure 4.19 highlights two weak cells. Hand-over-mouth precision is only 0.54 due to look-alikes, and one held-out driver is detected only half of the time, which is why we mine hard negatives and always evaluate on unseen drivers.

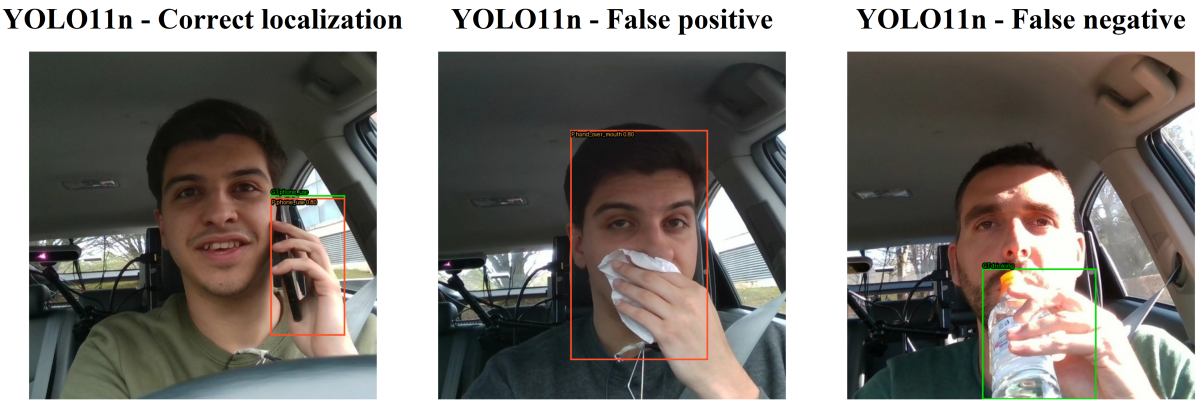


Figure 4.18: YOLO11n on held-out DMD drivers: correct detection, false positive and missed detection (from our benchmark [78])

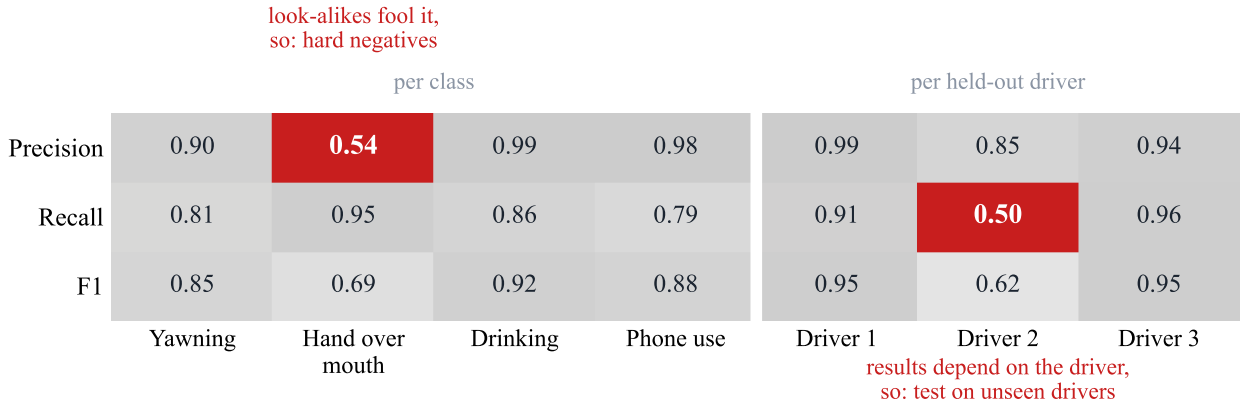


Figure 4.19: Where YOLO11n is weak: precision, recall and F1 per class and per held-out DMD driver, mean of three seeds [78]

False alarms and cropping. Both studies are presented in Chapter 1 (Figure 1.3): 40% hard-mined negatives cut false positives by 71% [50], and cropping to the driver preserves accuracy down to 320 px.

Compression and cutoffs. Structured pruning of up to 50% kept the per-class AP_{50} of YOLO11n within about two points (mAP_{50} from 0.933 to 0.928), yet at a fixed cutoff of 0.5 its worst-class recall fell by 6.8 points. Cutoffs tuned on validation drivers met a recall floor on unseen test drivers in at most one of three training seeds, and INT8 quantization with TensorRT lowered mAP_{50} from 0.933 to 0.874 [75].

Temporal filtering. In our RGB-thermal detection study, a five-frame moving average reduced false alerts from 9.0 to 2.1 per minute compared with raw thresholding, at the cost of about 0.4 s of extra delay, while learned temporal models (GRU and Mamba) were not significantly better and ran up to 500 times slower [79].

Table 4.15: Preliminary studies and the design decision each one supports

Study	Data	Main result	Design decision
Detector benchmark [78]	DMD, 14 drivers, subject-disjoint	YOLO11n best mAP_{50} (0.918) and speed of three nano detectors	YOLO11n selected
Negatives and hard negatives [50]	15,723 frames	40% hard-mined negatives: 24.4 → 7.1 false positives per 1000 frames	Hard-negative round (FS-12)
Pruning and quantization [75]	DMD, our four classes	AP kept up to 50% pruning, cutoffs fail across drivers and INT8 costs 5.9 mAP_{50} points	Per-driver calibration (FS-19), comparing PTQ and QAT
Temporal filtering [79]	RGB-thermal video	Moving average: 9.0 → 2.1 false alerts per minute	Simple time-based EMA (FS-04)
RoI cropping (Chapter 1)	Public DMS data	Crop keeps accuracy down to 320 px, ≤ 3 ms	Fixed crop, 320 px CPU fallback

4.3.6.3 What the results tell us

Together, the studies support YOLO11n, hard-negative mining and a simple time-based average. Most importantly, a cutoff tuned on some drivers does not hold for others, forming the strongest argument for per-driver calibration. The main risks are the INT8 accuracy drop on the NPU and how well results on public data transfer to our own recordings of the four authors.

4.4 Real-World Considerations

4.4.1 Environmental impact

The device adds a small amount of e-waste in the Pi, the HAT, the camera, the hub and the display, but contains no battery to dispose of, and at about 16 W it consumes less energy than a car's headlights. Training is the largest energy cost, which we limit by fine-tuning a small pre-trained model rather than training from scratch, and by running compression only once. The server stores only small event records, keeping its energy use minimal. At the end of the project, the parts are returned to the university lab for reuse.

4.4.2 Ethical and social issues in the UAE context

4.4.2.1 Privacy

Our own dataset records only the four authors (Section 5.3), so no outside participants are involved. For a product, every driver would give consent under the UAE PDPL [68], with GDPR [69] as a reference, and a "delete my data" button on the dashboard remains a Future item.

4.4.2.2 Social and cultural perception in the UAE

An in-car camera can feel intrusive, especially for families and for women who may prefer not to be filmed. Keeping all video on the device and never uploading it is our main answer, together with a visible camera that is never hidden. Fleet employees, such as taxi drivers, may fear the system being used to punish them, so our design gives the data to the driver first and shares only events, never video. Voice messages in both Arabic and English make the system feel local and clear, and since phone use costs 800 AED and four black points in the UAE [49], drivers have a personal reason to welcome a gentle reminder.

4.4.2.3 Fairness and responsible use

Detection quality may differ for drivers with glasses, headscarves, beards or different skin tones. With only the four authors in our own dataset, we cannot measure this ourselves, so we fine-tune on the 14 drivers of DMD first and report results per driver, leaving a larger and more varied test set as a Future item. Alert fatigue is itself a safety risk, which is precisely why the cool-down, calibration and tiered alerts exist. The system is designed to assist and explain rather than punish, giving tips instead of scores, and it is never presented as a replacement for the driver's attention.

4.4.3 Economic analysis

4.4.3.1 Cost to produce

The prototype costs 2725 AED, of which the 250 AED wheel serves the demonstration rig and the haptic prototype. A single unit costs about 2160 AED without the display, and a custom board with the same NPU could bring this below 800 AED at volume. Cloud costs stay small, as only events are uploaded (Table 4.16).

Table 4.16: Unit cost and cloud cost

Quantity	**BOM per unit (AED)**	**Cloud cost / month**
1 unit, without display	About 2160	Free tier
1 unit, with display	About 2475	Free tier
Volume (1000+ units, custom board)	Below 800 (estimate)	About 1–2 AED per driver

4.4.3.2 What makes our product competitive

Unlike the products of Section 2.3, our device is a one-off purchase that works in any car, gives the feedback to the driver, keeps video on the device, reports false alarms per hour and calibrates per driver. In a production vehicle, each prototype part has an existing equivalent: an NIR camera in

the cluster, a driver-assistance computer with an NPU, an ambient LED strip, the cabin speakers and a haptic steering wheel. Table 4.17 summarises the strengths and risks.

Table 4.17: SWOT analysis

Strengths	**Weaknesses**
Low cost with no subscription, video never leaves the car, explained and tiered alerts, per-driver calibration and open evaluation	RGB only and weak at night, four classes only, and a bench prototype not certified for vehicles
Opportunities	**Threats**
EU GSR mandates raise awareness, UAE fleets (taxis, RTA), a retrofit market for older cars and an Arabic-first interface	Factory-fitted systems in new cars, the AGPL licence if commercialised and driver acceptance of in-car cameras

Chapter 5 Research Methodology

5.1 Design methodology: from requirements to final design

We follow an iterative engineering process: specifying numbered requirements (Section 4.1), comparing alternatives against weighted criteria (Section 4.2), then prototyping, measuring and refining. The model follows its own inner loop of collecting clips, training, evaluating on the validation split, recording hard negatives that target the errors, and retraining. Only once a model meets the targets on the Pi do we freeze it and run the test set, a single time (Figure 5.1).

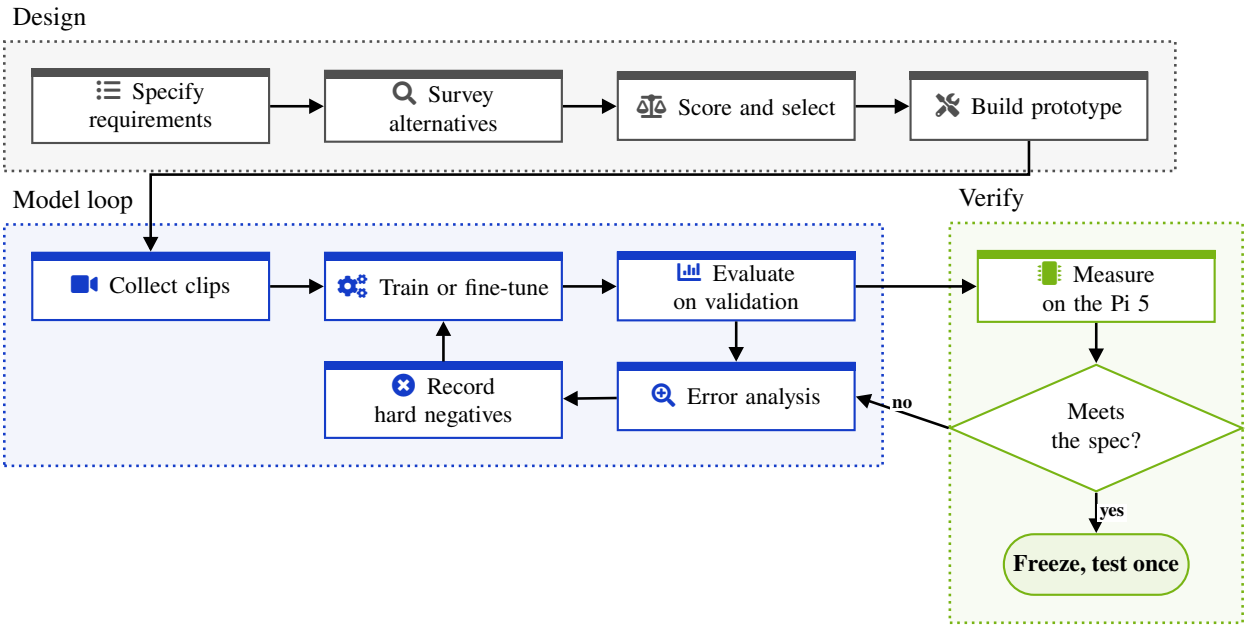


Figure 5.1: Design process with the repeating model loop

5.2 Skills, techniques, models and tools needed

Table 5.1 lists the skills and tools the project requires, all of which are free or open source.

Table 5.1: Skills, tools and licences

Area	Skills and tools	Licence or cost
Model	Python, PyTorch, Ultralytics YOLO11, transfer learning, structured pruning, PTQ and QAT, TensorBoard	AGPL-3.0 for Ultralytics, which needs open-sourcing or an enterprise licence if the product is sold
Data	CVAT annotation, OpenCV, the subject-disjoint protocol	Free
Embedded	Raspberry Pi OS, Hailo runtime and Dataflow Compiler, GPIO, boot services, one small C++ library	Free with a Hailo developer account
Server and web	SQLite, FastAPI, PostgreSQL, React, free hosting tiers, OpenStreetMap Overpass API [70]	Free, maps under ODbL with attribution
Design	Git and GitHub, draw.io, Mermaid, TikZ, Blender, MATLAB for the alert sounds	Free or university licence
Measurement	Multimeter, inline USB-C power meter, infrared thermometer, phone filming at 240 FPS	Lab stock

5.3 Dataset: Collection, Labelling, Splits and Hard Negatives

5.3.1 Dataset goals and overview

The dataset must teach the detector our four classes from our own camera view, while allowing us to measure how well the model generalises to people it has never seen. Besides the four positive classes, every frame without a cue serves as a negative, with hard negatives added in a second round. Table 5.2 gives the plan.

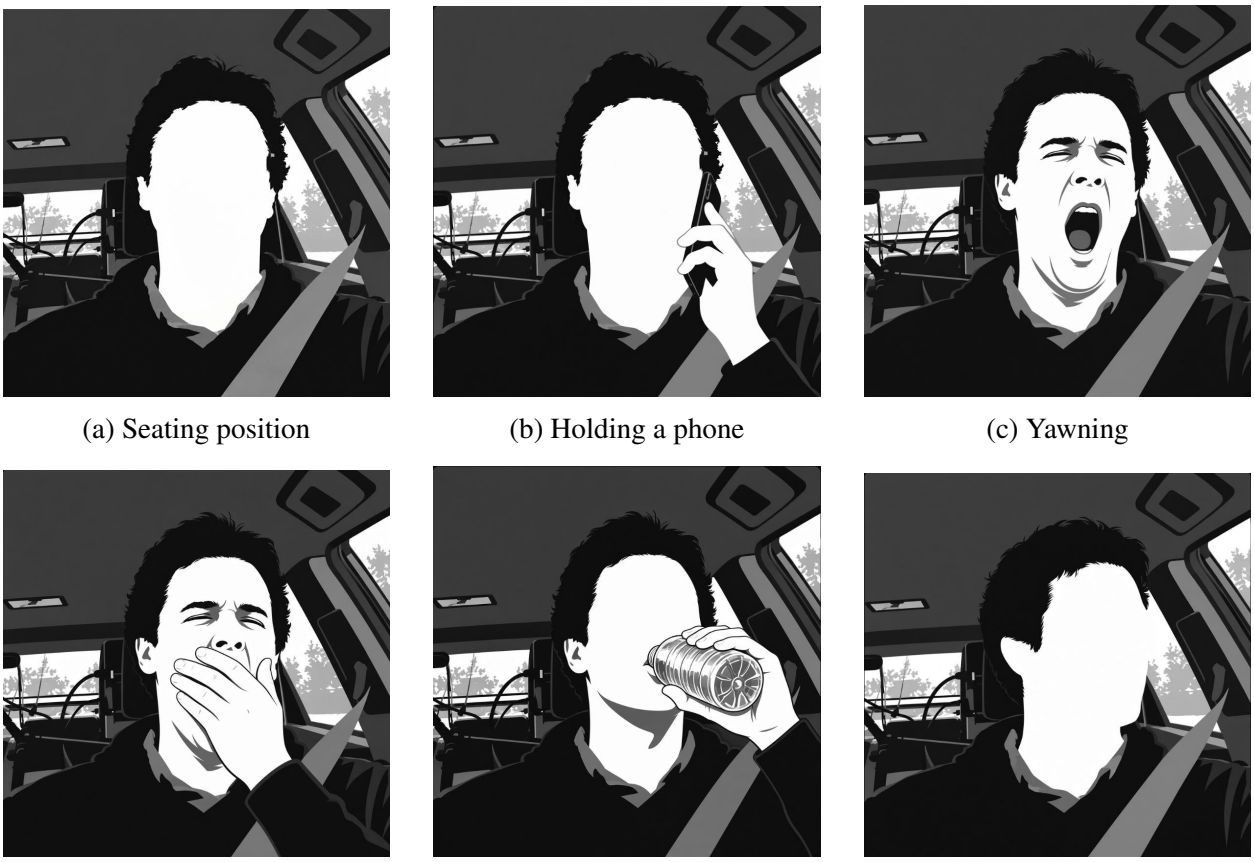
Table 5.2: Planned dataset

Item	Plan
Subjects	The four authors (S01 to S04), with appearance variety coming from pretraining on the 14 drivers of DMD
Sessions	Several 10-minute sessions per subject, in the afternoon and in the evening, each with a 20 s neutral baseline
Hours of video	About 2.7 hours (at least four sessions per subject), plus about 0.5 hours of hard-negative clips
Frame counts	Labelled at 5 FPS: about 48,000 frames, of which roughly 15–20% contain a cue
Classes	Yawning, hand-over-mouth, drinking, phone use, plus negatives (including looking away)

5.3.2 Collection protocol

5.3.2.1 Recording setup and conditions

The camera is mounted in the target in-car position (Figure 4.8). Each subject sits in a parked car with the engine off and records several 10-minute RGB sessions at 1080p and 30 FPS, in the afternoon and in the evening. Following the guideline in Figure 5.2, the subject performs each behaviour several times at random moments, separated by periods of normal posture. Looking away is recorded deliberately and labelled as a negative, so that mirror checks teach the model what not to alert on.



(a) Seating position (b) Holding a phone (c) Yawning

(d) Covering a yawn (e) Drinking (f) Looking away (negative)

Figure 5.2: Recording guideline given to every subject

5.3.2.2 Pilot sessions and what they taught us

Before the C922 arrived, three of the authors (P1 to P3) recorded four pilot sessions in three cars, by day and by night, using phones in different mounts (Table 5.3 and Figure 5.3). These clips are neither labelled nor part of the dataset, as we used them only to check the protocol and the camera

mount by eye. Two of them were filmed upside down, showing that the frame grabber needs a 180° rotation setting.

Table 5.3: Pilot sessions recorded in November 2026

Clip	Subject	Light	Camera and mount	Format	Length
C1	P1	Night, parked, blue cabin light	iPhone low on the dash, upside down	3840×2160, 14 FPS	92 s
C2	P2	Night, warm dome light	Phone on the dash, portrait	478×850, 30 FPS	40 s
C3	P2	Day, sun, seatbelt on	Phone on the dash, portrait	478×850, 30 FPS	92 s
C4	P3	Night, red and green street light	Phone below the wheel, upside down	1024×576, 30 FPS	29 s

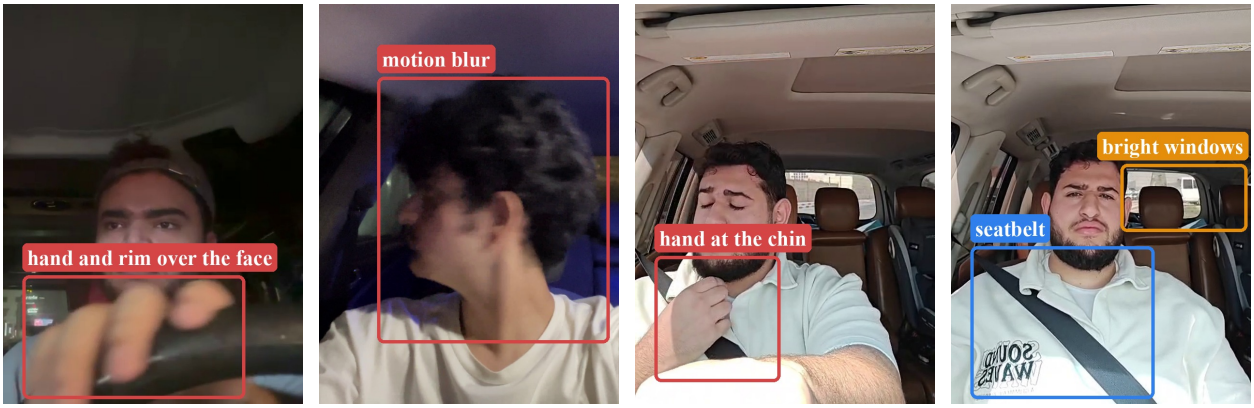
(a) C1: night, low on the dash (b) C2: night, on the dash (c) C3: day, on the dash (d) C4: night, below the wheel

Figure 5.3: One frame from each pilot session, turned upright where needed

Figure 5.4 and Table 5.4 summarise what the sessions taught us, with each lesson visible in its frame. Figure 5.5 follows one drive in C4, where within 25 s the same face turns green, red, evenly lit and dark. An RGB camera captures all of this variation, which is why training uses hue and brightness augmentation (Table 4.10) and why an IR camera remains the first Future item.

5.3.2.3 Subjects and data handling

The subjects are the four authors, so no outside participants are recorded and each author has agreed to appear in the dataset and in this report. Only an anonymised ID (S01 to S04) appears in file names and labels, and the videos stay on an encrypted university drive with access limited to the team and the supervisor. Since we plan to release the dataset publicly, each author may still withdraw their clips before the release.



(a) Below the wheel, the hand and rim hide the face (b) A fast head turn blurs at 14 FPS (c) A hand at the chin with the eyes shut (d) Bright windows and a seatbelt in daylight

Figure 5.4: What the pilot sessions showed, with the problem marked in each frame (Table 5.4)

Table 5.4: What the pilot sessions taught us

Frame	What we saw	What we do
(a)	Below the wheel, the rim and the hand cross the face	Keep the dash-top mount above the column (Section 4.3.3)
(b)	A fast head turn is blurred at 14 FPS	Record at 30 FPS with the C922, and keep the 3 s rule so one blurred frame never reads as a blocked camera
(c)	A hand at the chin and neck, with the eyes shut and the head turned	Keep the chin row in the hard-negative list (Table 5.6), and test the eye rule with the head turned and down (T-20)
(d)	Bright rear windows behind the driver in daylight, and a seatbelt across the chest	Set the camera exposure for the face, not the whole frame, and treat the seatbelt as part of a normal pose

(a) 3.0 s: green light (b) 4.2 s: red light (c) 8.9 s: even light (d) 24.3 s: almost dark

Figure 5.5: One night drive (C4): street light changes the colour and brightness of the same face within 25 s

5.3.2.4 Neutral baseline clips

Before each session, the subject sits normally for 20 s. These clips serve two purposes: benchmarking the new-driver calibration of Section 4.3.4.6 on people the model has not seen, and measuring each subject's open-eye EAR for the eye module.

5.3.3 Labelling

5.3.3.1 Class definitions and annotation rules

A positive must be visible for at least 0.5 s, with the box covering the region that defines the class. Look-alikes are deliberately labelled as negatives, and head orientation is never labelled, so mirror checks and looking-away segments always remain negatives. Table 5.5 gives the rules.

Table 5.5: Class definitions and annotation rules

Class	**Positive definition**	**Look-alikes (negative)**	**Box rule**
Yawning	Mouth wide open with jaw dropped, at least 0.5 s	Talking, singing, laughing	Mouth and chin
Hand-over-mouth	Palm or back of hand covering the mouth	Touching nose or chin, adjusting headscarf	Hand and mouth region
Drinking	Bottle or cup at the lips	Holding a cup below the chin, eating	Container and mouth
Phone use	Phone at the ear or held in view and looked at	Hand at the cheek, phone in a holder	Phone and hand

5.3.3.2 Annotation tool and quality checks

Labels are drawn in CVAT on frames sampled at 5 FPS, with interpolation between key frames to speed up long cues. For the first subject, two annotators label each video independently, measuring agreement as the share of boxes that match at an IoU of 0.5 or more and discussing disagreements to refine the rules in Table 5.5. From then on, one annotator labels each video and the second checks a random 10% of frames, relabelling any video where more than 5% of the checked frames disagree.

5.3.4 Subject-disjoint evaluation with four subjects

With only four subjects, a fixed split would leave a single person in the test set, so we use leave-one-subject-out testing instead. The model is first fine-tuned on the 14 drivers of DMD, then on our data in four folds, where each fold holds out one subject for testing and one for validation and

trains on the remaining two, so that

$$\mathcal{S}_{\text{train}} \cap \mathcal{S}_{\text{val}} = \mathcal{S}_{\text{train}} \cap \mathcal{S}_{\text{test}} = \mathcal{S}_{\text{val}} \cap \mathcal{S}_{\text{test}} = \varnothing \quad \text{in every fold.} \tag{5.1}$$

Every session, baseline clip and hard-negative clip of a subject stays with that subject. Results are reported as the mean and spread over the four folds, ensuring every subject is tested exactly once by a model that never saw them. Figure 5.6 shows the folds.

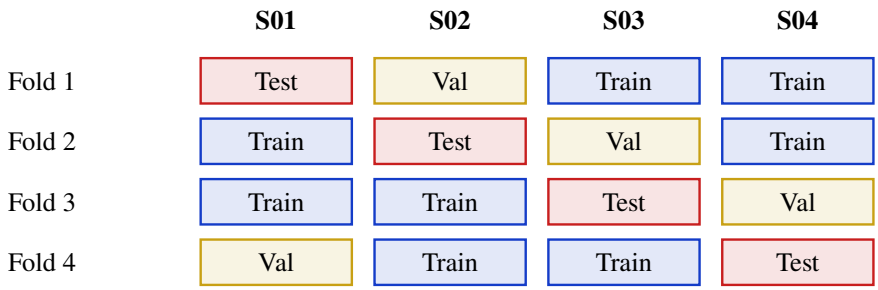


Figure 5.6: Leave-one-subject-out folds: every subject is tested once by a model that never saw them

5.3.5 Hard-negative dataset

After the first model is trained, we run it at a low confidence on the training and validation clips and collect the frames it wrongly flags. The same subjects then record a short session deliberately acting these look-alikes (Table 5.6). Only negative frames are added, with each subject kept in their own split, and false alarms per hour are compared before and after, using the 40% hard-mined ratio from Chapter 1.

Table 5.6: Planned hard negatives

Look-alike behaviour	**Class it may fool**	**Frames added (target)**
Talking, singing, laughing	Yawning	About 800
Rubbing nose or chin, adjusting headscarf or glasses	Hand-over-mouth	About 800
Holding a cup below the chin, eating a snack	Drinking	About 600
Hand resting on the cheek, touching the ear	Phone use	About 600
Phone in a dashboard holder, not held	Phone use	About 400

5.4 How We Will Prove It Works

5.4.1 What will be verified: each specification matched to a test

Every specification of Tables 1.1 and 1.2 maps to at least one verification method: test (measured against a pass criterion), analysis (calculation or review of logs), inspection (checking the build or the code) or demonstration (showing the feature working with a user). Test T-*xx* verifies specification FS-*xx*, and Table 5.7 lists the main feature tests.

5.4.2 Test procedures

5.4.2.1 Model tests on the subject-disjoint test set

The final model is tested leave-one-subject-out on our four subjects, as well as on the three held-out DMD drivers. We report per-class precision, recall, F1 and mAP with a confusion matrix, computed per frame, while false alarms per hour are computed on the full pipeline output over the neutral and hard-negative clips, before and after hard-negative retraining. The test split is never used to tune cutoffs or τ.

5.4.2.2 Performance tests on the Raspberry Pi 5

Each model variant runs on the Pi for 30 minutes on a recorded test clip in a room at about 30 °C, with the case closed and the cooler running. We log FPS, end-to-end latency (mean and 95th percentile), SoC temperature and any throttling flags reported by the firmware, alongside the accuracy of the same variant on the validation set. The variants are the baseline at 640 and 320 px on the CPU, the baseline on the Hailo NPU, and 20% and 30% pruned models on the NPU, each with PTQ and QAT.

5.4.2.3 Feature tests

Each feature has a simple, repeatable test (Table 5.7). Eye closure is tested on the demonstration rig of Section 4.3.3.2, including closures with the head down, since these were confused on the DMD footage. All feature tests run on the same rig, always with subjects whose clips were not used for training.

Table 5.7: Feature tests

Test ID	Feature	Procedure	Pass criterion
T-04	EMA filter	Replay one clip at 10, 15 and 30 FPS	Confirmation times agree within one frame
T-05	Light and voice alert	Film the bar at 240 FPS and listen to the message	≤ 3 flashes/s and the message plays after every confirmed event
T-06	Cool-down	Two same-class cues 10 s apart	Second cue not alerted, both logged
T-07	Mirror checks	20 scripted mirror checks	No alert
T-15	Escalation	Hold a cue for 15 s	Tier 2 at 5 s, Tier 3 at 10 s (± 0.2 s)
T-16	Explained messages	Trigger each class	Message names class and count or duration
T-17	Camera-blind check	Cover the lens or shine a torch, 10 trials each	Announcement within 3–3.5 s in all trials
T-18	Crop tracking	Same clips, fixed vs tracked	Tracked recall $\geq$ fixed, no extra false alarms
T-19	Calibration	Test subjects' baselines	False alarms/h not higher than defaults
T-20	Eye closure	Acted closures of 1, 1.5 and 2 s, and 2 and 4 s with the head down	Alert at 1.5 s and 2 s, none at 1 s, and with the head down none at 2 s and alert at 3 s
T-24	Display	Five users perform setup and read status	All complete setup unaided and read the status in under 2 s
T-26	Rest-stop suggestion	Simulate high alert rate at a known location	Nearest PoI named within 5 s
T-21	Power-cut logging	A smart relay cuts power 50 times during writes	Integrity check passes, no committed event lost
T-22	Secure upload	Upload to a server with a bad certificate, and drop Wi-Fi during an upload	Upload refused, no event lost or duplicated, no readable traffic
–	Endurance	Two-hour continuous run	No memory leak, throttling or missed frames

5.4.3 Expected results and pass/fail criteria

Required specifications are must-pass, so a failure is fixed before any Desired feature is added. Desired specifications are aims, and a miss is reported with its cause rather than blocking the project. Table 5.8 gives the target and minimum acceptable value for the key specifications.

Table 5.8: Expected results and pass/fail criteria

Spec	Target	Minimum acceptable	Type	Result (SDP 2)
FS-01	30 FPS, ≤ 60 ms	25 FPS, ≤ 100 ms	Must pass	
FS-03	Recall ≥ 0.95	Recall ≥ 0.90	Must pass	
FS-04	Agreement within 1 frame	Within 2 frames	Must pass	
FS-05	≤ 3 flashes/s	≤ 3 flashes/s	Must pass	
FS-06	No repeat in 30 s	No repeat in 30 s	Must pass	
System	≤ 3 false alarms/h	< 10 false alarms/h	Must pass	
FS-11	≥ 40 FPS after compression	Recall drop ≤ 2 points	Aim	
FS-17	Detected in 3 s	Detected in 4 s	Aim	
FS-19	30% fewer false alarms	Not more than default	Aim	
FS-20	Alert at 1.5 s	Alert by 2 s	Aim	
FS-21	50/50 power cuts clean	50/50 power cuts clean	Aim	

5.5 Project Plan: Parts, Budget, Timeline and Risks

5.5.1 Hardware parts list

Table 5.9 lists every item in our approved purchase request, all sourced from Amazon.ae for fast delivery in the UAE, with each supplier entry linking to the product page. The MOSFET and resistors for the light-bar driver come from the department lab stock at no cost. The AI HAT+ moved from optional to Required once our compute estimate showed that the CPU alone cannot reach the target FPS, and the PXN wheel serves both the demonstration rig and the prototype of the Future steering-wheel alert.

5.5.2 Project budget

The department funds the approved purchase request against a cap of 5000 AED. With a 15% contingency for spares, a car power adapter and cables, the total comes to about 3134 AED (Table 5.10).

5.5.3 Timeline and resource plan for SDP2

5.5.3.1 Milestones and Gantt chart

Required features come first and data collection starts early, as recording, labelling and the hard-negative round form the bottleneck. The Required build is frozen by week 8, after which the Desired features follow in priority order with a buffer for the Eid al-Fitr break, testing runs in weeks 13 to 15, and the final two weeks are reserved for the report, poster and presentation (Figure 5.7).

Table 5.9: Hardware parts list and estimated budget

#	Item	Supplier	Qty	Unit (AED)	Total	Status
1	Raspberry Pi 5 (8 GB)	Amazon.ae	1	845	845	Required
2	Raspberry Pi AI HAT+	Amazon.ae	1	391	391	Required
3	Logitech C922 webcam	Amazon.ae	1	255	255	Required
4	Creative Pebble V3 speakers	Amazon.ae	1	177	177	Required
5	27 W USB-C power supply	Amazon.ae	1	112	112	Required
6	Raspberry Pi 5 Active Cooler	Amazon.ae	1	64	64	Required
7	SanDisk Extreme Pro 128 GB	Amazon.ae	1	136	136	Required
8	Metal case	Amazon.ae	1	86	86	Required
9	UGREEN micro-USB cable	Amazon.ae	1	22	22	Desired
10	Hosyond 7-inch touch display	Amazon.ae	1	245	245	Desired
11	KABCON tablet holder	Amazon.ae	1	44	44	Desired
12	UGREEN USB 3.0 hub	Amazon.ae	1	28	28	Required
13	RGB light bar	Amazon.ae	1	70	70	Required
14	PXN steering wheel	Amazon.ae	1	250	250	Future (prototype)
–	Logic-level MOSFET, 220 Ω and 10 kΩ resistors, jumpers	Lab stock	–	0	0	Required
				Grand total	**2725**	

Table 5.10: Project budget (AED)

Category	Required	Desired	Future	Total
Hardware	2164	311	250	2725
Software and services	0	0	0	0
Contingency (15%)	325	47	37	409
Total	**2489**	**358**	**287**	**3134**

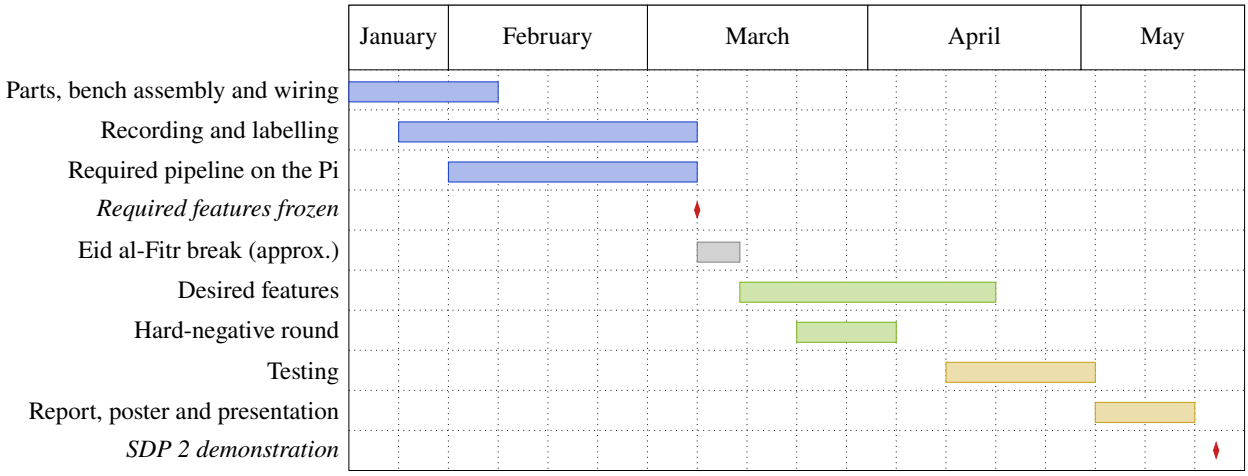


Figure 5.7: Gantt chart for SDP 2

5.5.3.2 Task assignment for each team member

Each task has one owner and one support member who can take over (Table 5.11).

Table 5.11: Task assignment

Task	**Owner**	**Support**	**Due (approx.)**
Dataset collection and labelling	Oumar	Mohamed	7 Mar 2027
Training, pruning, hard negatives	Oumar	Abdullah	4 Apr 2027
Bench assembly, wiring, power	Abdullah	Faris	7 Feb 2027
Hailo compile and on-device pipeline	Abdullah	Oumar	21 Feb 2027
EMA, decision logic, alerts, logging	Abdullah	Mohamed	7 Mar 2027
Upload API, database, dashboard	Faris	Oumar	11 Apr 2027
Display UI, calibration screens, voice messages	Mohamed	Faris	3 Apr 2027
Eye module and crop tracking	Oumar	Abdullah	18 Apr 2027
Testing and user trials	Mohamed	All	2 May 2027
Final report and poster	All	All	16 May 2027

5.5.4 Risks and backup plans

Table 5.12 lists each risk with its likelihood, impact and mitigation. The largest risks are delays in data collection and poor generalisation from only four subjects, which we address by starting early, pre-training on public data and calibrating per driver.

Table 5.12: Risks and backup plans

ID	Risk	Likelihood	Impact	Mitigation
R1	Data-collection delays	Likely	High	Four pilot sessions are already recorded (Section 5.3.2.2). Book slots in week 1, record two sessions per day and fall back on DMD and YawDD for missing classes
R2	Poor generalisation to new subjects	Likely	High	Only four subjects: pre-train on 14 DMD drivers, leave-one-subject-out testing, per-driver calibration, hard negatives
R3	Pi too slow or too hot	Possible	Moderate	AI HAT+, 320 px CPU fallback, active cooler, vented case
R4	Too few hard negatives	Possible	Low	Script look-alikes in advance and mine false positives from public sets
R5	Demonstration room differs from the recorded cars	Possible	Moderate	Per-driver calibration at the start of each demonstration, brightness and hue augmentation, and the blind check
R6	Supply delays	Possible	Moderate	Order in week 1, with local alternatives for every part
R7	Hailo compile issues	Unlikely	Low	Use Hailo Model Zoo configs for YOLO11n, with a CPU fallback
R8	Map and PoI service access	Likely	Low	Overpass API needs no key but live queries timed out in our tests, so a saved list of 363 places answers within the limit
R9	Eye alarm false alerts	Likely	High	Record real closures at full frame rate, retune on the C922, and keep the alarm out of the frozen build until it meets the target

Chapter 6 Implementation and Results So Far

Since the SDP 2 hardware has not yet arrived, we built and tested everything that does not depend on it. This chapter reports what exists today, how it was checked and what was measured. Every number here comes from a run on a development PC rather than on the Raspberry Pi 5, and we keep the two apart throughout, leaving the Pi results for SDP 2.

6.1 What has been built

The full software of Chapter 4 now exists as one repository of 86 source files, with about 5500 lines of Python, 1300 lines of web code and one small C++ library, running from a camera or a video file through to the web dashboard. Table 6.1 lists the parts. The modules follow the class diagram of Figure 4.9, and the detector backend is selected in one place, so the same pipeline runs on PyTorch, ONNX, NCNN or, once the AI HAT+ arrives, the Hailo runtime.

The detector in every run below is our YOLO11n trained on DMD in the benchmark of Chapter 4, rather than a model trained on our own subjects, since that dataset has not yet been recorded.

6.2 How it was tested

Testing relied on two kinds of evidence: automated scripts that check the logic against the test plan of Section 5.4, and a replay that exercises the whole pipeline on real footage. The replay covers 648 s of held-out DMD driver `subject_05`. Since the DMD frames we hold are sampled at 1 FPS, each frame was repeated ten times to form a 10 FPS clip with a matching label file. This suffices to exercise the full system, although event times on this clip are only accurate to about a second.

All 194 automated checks pass across the eight test scripts of Table 6.2. These are software checks, where the light bar and audio are replaced by recording stand-ins and power cuts are simulated by killing the writing process.

Table 6.1: Software built so far

Part	**What it contains**	**Technology**
Edge pipeline	Capture, crop, blind check, detector, EMA, decision state machine, alerts, calibration, eye module, crop tracking	Python, C++, OpenCV, Ultralytics, MediaPipe, YuNet
Event log	Driver, session, event, calibration and tip tables, integrity check at boot, recovery of unfinished sessions	SQLite in WAL mode
Driver display	PIN login, calibration, pre-trip check, status, settings, rest-stop card	Flask, HTML, 1024×600
Alert assets	3 alarm sounds from MATLAB, 40 voice messages in English and Arabic (80 clips)	WAV files, neural text-to-speech
Trip assistant	Nearest rest stop from OpenStreetMap, with a saved list of 363 places as a fallback	Overpass API
Upload and server	TLS 1.3 uploader with device key, retry and duplicate handling, login, statistics, tips	FastAPI, PostgreSQL or SQLite
Dashboard	Overview, trips, statistics, tips and privacy pages, with hand-drawn charts	React, Vite
Dataset tools	Recording guide, frame extraction, subject-disjoint folds, annotator agreement, hard-negative mining	Python
Model tools	Training, evaluation, structured pruning, ONNX and NCNN export, Hailo compile script, Pi benchmark	Ultralytics, torch-pruning
Deployment	Pi setup script, boot service, kiosk start	Bash, systemd

Table 6.2: Automated software checks

Script	**What it covers**	**Passed**
Decision	EMA at 10, 15 and 30 FPS, confirmation, cool-down, tiers, eye rules	29/29
Logger	Schema, PIN storage, 50 process kills, integrity recovery	16/16
Vision	Detector, crop, blind check on synthetic and real frames, C++ against OpenCV	25/25
Alerts	Flash limit, sound and voice selection, clip coverage, muted cues	14/14
Eye and calibration	Calibration formula, EAR, PERCLOS, closure timing, look-down rules, crop tracking on a real face	42/42
Display	PIN flow, calibration, pre-trip check, display states	26/26
Upload	TLS refusal of bad certificates, device key, retry, duplicates, per-driver access, tips	29/29
Tools	Subject-disjoint folds, agreement measure, dataset helpers	13/13
Total		**194/194**

6.3 Results so far

6.3.1 Full pipeline replay

Figure 6.1 and Table 6.3 present the replay. All four phone cues were caught and escalated, with tier times landing exactly at 5 s and 10 s after the cutoff crossing. Two of the four drinking cues were missed, at 148 s and 227 s, both detector misses on short cues from a driver the model never

saw, and no detector alert fell outside a labelled cue. One result points to a design issue rather than a model issue: during the last long phone call, a brief drop in detection counted as the end of the cue, so the rest of the call fell inside the 30 s cool-down and stayed silent.

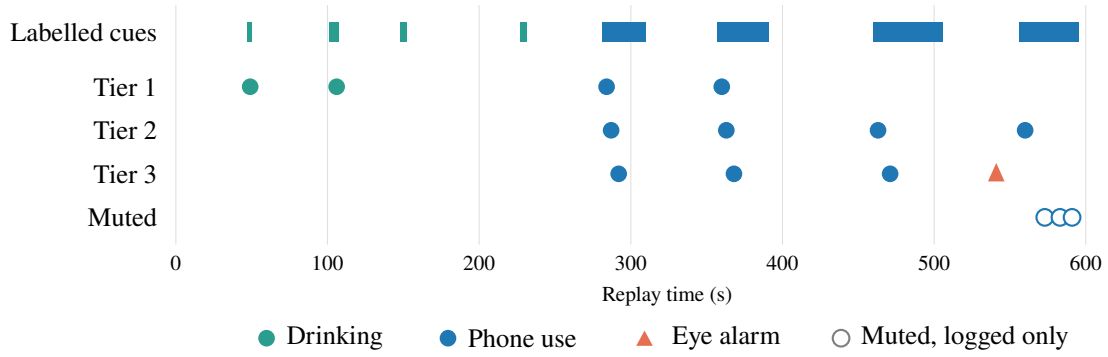


Figure 6.1: Replay of held-out DMD driver `subject_05`: labelled cues against the alerts the system raised

Table 6.3: Replay results on PC hardware

Measure	**Result**	**Target**
Phone cues caught	4 of 4	Recall ≥ 0.90
Drinking cues caught	2 of 4	Recall ≥ 0.90
Tier 2 and Tier 3 timing	5.0 s and 10.0 s	±0.2 s
Detector alerts outside labelled cues	0 in 648 s	≤ 3 per hour
Eye alarms on the same clip	1 (26 with the first rule)	≤ 3 per hour
Latency, RTX 4060, no eye module	37 ms	Pi target: ≤ 100 ms
Latency, CPU, no eye module	111.4 ms mean, 116.1 ms p95	Pi target: ≤ 100 ms
Latency, CPU, with eye module	143.1 ms mean, 181.0 ms p95	Pi target: ≤ 100 ms

These numbers serve as a working check rather than an evaluation. One driver and eight cues cannot estimate recall, and 648 s cannot estimate a false alarm rate, so the final figures will come from the leave-one-subject-out test of Section 5.4. The PC latencies also say nothing about the Pi, where detection runs on the NPU, and they overstate the work done per frame, since a replay keeps every frame and the figures include the time a frame waits in a queue. Timed directly, the detector and eye module together take about 71 ms per frame on the CPU.

6.3.2 Logging, upload, compression and other checks

Table 6.4 lists the other results.

Table 6.4: Other results on the PC

Item	Result	Still open
Logging	50 process kills left the database intact, with no committed event lost	Real relay test, because `synchronous=NORMAL` can lose the last commits in a power cut
Upload and dashboard	41 replay events uploaded over TLS 1.3 and shown in the dashboard. Bad certificates refused, no duplicates, each login sees only its own data	PostgreSQL, Docker and hosting not run
Pruning	A 20% target removed 24.9% of the parameters and dropped mAP_{50} to 0 until fine-tuning, and 8 epochs recovered it	Full comparison with the acceptance rule
Export	ONNX and NCNN match PyTorch (confidence 0.818)	FP16, INT8, QAT and the Hailo compile
Trip assistant	The saved list of 363 places answered in 4.31 s after the live query timed out at 4 s	A measured live answer
Blind check	Brightness 59 to 107 and sharpness 126 to 365 on DMD frames, with no false “camera blocked”	Limits tuned on the C922

6.3.3 Eye closure

The first eye rule, an EAR below 75% of the open-eye value for 1.5 s, did not survive real footage. Across all 15,723 frames of the 14 alert DMD drivers, it fired about 100 times per hour against a target of 3. In a sample of 192 flagged frames, about a third showed a visible iris, during a smile, a squint or a slight downward look, while about 40% were a real lid closure with the head looking down at the lap or the phone. From a dashboard camera, a downward look and a closed eye can appear identical in a single frame.

The rule now has three changes, all reflected in Table 4.11. The EAR must fall below 50% of the open-eye value, MediaPipe’s eye-blink score must also exceed 0.50, and when the head is more than 8° below the driver’s rest pitch, the alarm waits 3 s instead of 1.5 s. This last step only delays the alarm and never mutes it, as a driver nodding off also has the head down. Head-down time is also excluded from PERCLOS, and the rest pitch is measured during calibration alongside the open-eye EAR.

We tuned these values on drivers 1 to 9 and checked them on drivers 10 to 14. Figure 6.2 shows a drop of about 80% that holds on the unseen drivers. At 1 FPS the rate can only be bounded, so the figure counts runs of two frames, which overcount, and runs of three frames, which undercount, and even the lower figure remains above the target of 3 per hour. On the replay clip, the eye alarms fell from 26 to 1.

How many real closures the new rule misses remains unknown, as we have no labelled closed-eye footage, and real closures recorded at full frame rate on the C922 are needed before these values can be trusted.

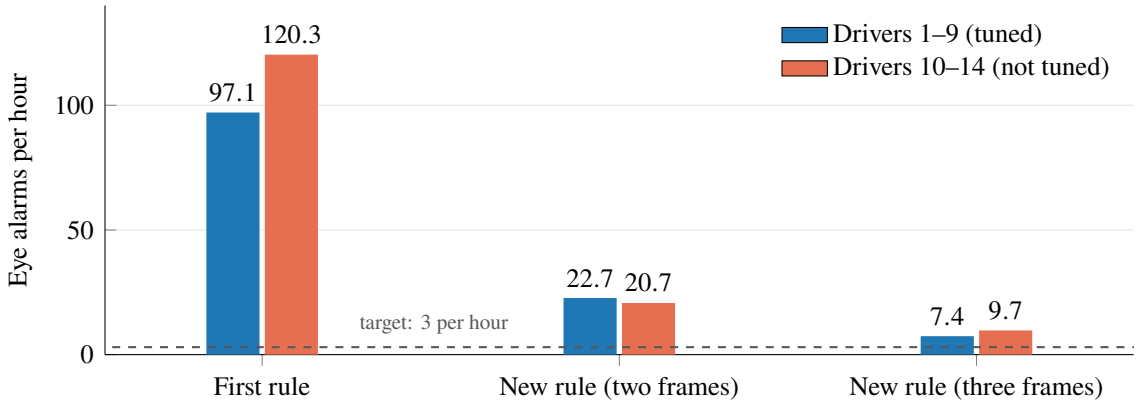


Figure 6.2: Eye alarms per hour on 14 alert DMD drivers, before and after the change

6.3.4 Where the time goes

We timed each stage on the replay clip to estimate what a C++ rewrite would gain (Table 6.5). Most of the Python only coordinates native libraries, so a full rewrite would save very little, and the detector cost does not apply on the Pi, where the NPU takes over.

The one every-frame cost under our control was the blind check, which therefore became a small C++ library. It takes 0.32 ms against 3.0 ms for OpenCV and reaches the same decision on all 15,723 DMD frames, with the OpenCV version kept as a fallback when the library is not built.

Table 6.5: Time per frame by stage on the PC CPU

Stage	**Time**	**Where the work runs**
Detector, CPU path only	58.6 ms	Mostly native, 45.5 ms is the model
Eye module	About 11 ms	MediaPipe, native
Blind check	3.0 → 0.32 ms	Moved to C++
Video decode	1.2 ms	OpenCV, native
EMA and decision	About 0.002 ms	Python

6.4 Status of every specification

Table 6.6 gives the state of each specification today. “Tested” means the software behaviour was checked on a PC, and no entry implies that the specification has passed its final test on the Pi.

Table 6.6: Specification status at the end of SDP 1

Spec	**Feature**	**State**	**Evidence and what is left**
FS-01	Pi 5 deployment	Needs hardware	Setup, service and kiosk files written, never run on a Pi
FS-02	Camera	Partly tested	Tested on video files, the live C922 path is not run
FS-03	Detector	Partly tested	Runs with the DMD model, needs training on our subjects
FS-04	EMA	Tested	T-04 passes on synthetic scores
FS-05	Light and voice	Partly tested	Flash limit and clip selection pass, needs the real bar and speaker
FS-06	Cool-down	Tested	T-06 passes, but see the phone-call case above
FS-07	Mirror checks	Needs people	T-07 needs scripted mirror checks
FS-08	Logging	Tested	Replay events stored and read back
FS-09	On-device video	Tested	Upload contains event fields only
FS-10	Dataset	Pilot done	Four unlabelled pilot sessions with three of the authors (Section 5.3.2.2), the full dataset is still to be recorded
FS-11	Pruning and quantization	Partly tested	Pruning, fine-tuning, ONNX and NCNN ran, Hailo, QAT, FP16 and INT8 did not
FS-12	Hard negatives	Needs data	Miner ran but found none on a small sample, so it has yet to produce output
FS-13	Baseline clips	Needs camera	Recording tool written, not run
FS-14	Class alerts	Tested	Each class selects its own sound and pattern
FS-15	Tiers	Tested	T-15 passes, and exact on the replay
FS-16	Explained alerts	Tested	T-16 passes for every class
FS-17	Blind check	Partly tested	T-17 passes in software, limits need tuning on the C922
FS-18	Crop tracking	Partly tested	Follows a real face, stays off until T-18 shows it beats the fixed crop
FS-19	Calibration	Tested	Formula and redo rule pass, false-alarm benefit needs our subjects
FS-20	Eye closure	Known issue	Timing passes, but false alarms are above target (Section 6.3.3)
FS-21	Power-cut logging	Partly tested	50 process kills clean, relay test still needed
FS-22	Dashboard and sync	Tested	Real TLS upload to the dashboard on SQLite, PostgreSQL and hosting not run
FS-23	Tips	Tested	Rules checked against known histories
FS-24	Tablet	Partly tested	Screens checked in a browser, needs the 7-inch display and users
FS-25	Pre-trip check	Tested	Logic and rest advice pass, user trial open

Spec	Feature	State	Evidence and what is left
FS-26	Rest stops	Partly tested	T-26 passed on the saved list, live query not proven, phone location needs HTTPS on the Pi

6.5 What cannot be claimed yet

The results above show that the software works as designed on a PC, but not yet that the product meets its targets. Real FPS, latency, heat and power on the Pi are unmeasured, as is everything that needs the AI HAT+, the light bar, the speaker or the C922. Per-class recall and false alarms per hour on our own subjects require the dataset and the leave-one-subject-out test, while T-07, T-18, T-24 and all user trials require the demonstration rig of Section 4.3.3.2. Table 5.8 therefore stays empty until SDP 2. The four pilot sessions of Section 5.3.2.2 show real cabins, mounts and light, but they are neither labelled nor run through the model, so no detection result comes from them yet.

Chapter 7 Conclusion and Recommendations

7.1 Main findings

Nuisance alerts, whether true or false, lead drivers to distrust or switch off the very systems meant to keep them safe. We designed an edge-cloud DMS around the experience of the driver, in which a Raspberry Pi 5 with an AI HAT+ watches for yawning, hand-over-mouth, drinking and phone use, keeps all video on the device, and warns only when a behaviour is real, through alerts that explain themselves. On-device detection won our weighted comparison, and nuisance is controlled in layers: hard negatives, an EMA defined in seconds, confirmation times, a cool-down and per-driver calibration.

Our preliminary studies support YOLO11n, hard negatives and simple smoothing, while showing that cutoffs do not transfer between drivers. The full software now runs on a PC (Chapter 6), where all 194 checks pass and a replay of a held-out driver caught four of four phone cues and two of four drinking cues without any detector false alert. The most useful result was a failure: the planned eye rule fired about 100 times per hour, and our revised rule reduces this by about 80%, yet remains above target. Four pilot sessions with three of the authors in real cars confirmed the dash-top mount and showed how strongly night street light changes the face. No detection result so far comes from the Pi or from our own subjects.

7.2 Final impression: why this design is the right solution

Existing systems either detect well but remain in the lab, or are deployed but closed, cloud-dependent and designed for compliance rather than for the person driving. Our design sits in the gap between them, running a modern detector on affordable hardware, never sending video off the device, reporting the metric that matters to drivers and adapting to each driver. Above all, it treats every alert as a message to a tired person who simply wants to reach their destination, speaking rarely, clearly and only when it matters. We believe this is the kind of system a driver would choose to keep switched on.

7.3 Recommendations and what comes next

7.3.1 Next steps in SDP2

SDP 2 moves from building to proving. Once the parts arrive, we assemble the demonstration rig of Section 4.3.3.2, compile YOLO11n for the AI HAT+ and run the first Pi benchmarks of Section 5.4.2, while recording the dataset in parked cars, which is the longest task. The Required features are frozen by week 8, followed by the Desired features in priority order, with the eye module last, as it needs real closures recorded on the C922 (Section 6.3.3). Should time run short, the lowest-priority Desired features are dropped rather than delaying the Required build. The final demonstration takes place on the desk rig in a room, while in-car deployment remains beyond the scope of SDP 2.

7.3.2 Future extensions beyond SDP2

The Future items of Table 1.3 extend the system in three directions. The first adds sensing, where fusing RGB with IR would make the system work at night, building on our own RGB-thermal fusion study [79], while a second camera, wearables and access to the car's climate and audio would enrich what the system sees and how it responds. The second improves evaluation and adaptation, through in-car deployment, a robustness study across glare, low light, glasses, face coverings and different cars, a larger and more varied set of subjects, and alerts that stay quiet when vehicle signals show reversing, red lights or heavy traffic. The third covers interaction and privacy, with voice questions such as "how tired am I?", a real steering-wheel vibration alert based on our PXN prototype, and a delete-my-data button that gives drivers full control over their records.

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Chapter 8 Appendix

Appendix A gives the pseudocode of the core decision loop, Appendix B the design changes made during implementation, and Appendix C the submitted proposal.

Appendix A: Pseudocode of the decision loop

Algorithm 1 Per-frame filtering and alert decision

Require: cutoffs θ_c, time constants τ_c, confirm times T_c, cool-down $C = 30\,\text{s}$

$s_c \leftarrow 0$, $t_c^{\text{above}} \leftarrow \varnothing$, $t_c^{\text{mute}} \leftarrow -\infty$, $\text{tier}_c \leftarrow 0$ for all classes c
while session active **do**
  $f, t \leftarrow$ GrabFrame; $\Delta t \leftarrow t - t_{\text{prev}}$; $t_{\text{prev}} \leftarrow t$
  if Blocked(f) for 3 s or more **then**
    announce "camera blocked", log, reset all s_c and **continue**
  end if
  $x \leftarrow$ Detect(crop of f) ▷ max score per class
  for each class c **do**
    $\alpha \leftarrow 1 - e^{-\Delta t/\tau_c}$; $\quad s_c \leftarrow \alpha x_c + (1-\alpha) s_c$
    if $s_c \geq \theta_c$ **then**
      if $t_c^{\text{above}} = \varnothing$ **then** $t_c^{\text{above}} \leftarrow t$
      end if
      $d \leftarrow t - t_c^{\text{above}}$
      if $d \geq T_c$ **and** $t < t_c^{\text{mute}}$ **then**
        Log(t, c, x_c, d, tier 0) once ▷ muted, no alert
      else if $d \geq T_c$ **then**
        tier $\leftarrow$ Tier(c, d, recent events) ▷ 1, 2 or 3
        if tier > tier_c **then**
          Alert(c, tier, d); Log(t, c, x_c, d, tier); $\text{tier}_c \leftarrow$ tier
        end if
      end if
    else
      if $\text{tier}_c > 0$ **then** $t_c^{\text{mute}} \leftarrow t + C$; $\text{tier}_c \leftarrow 0$
      end if ▷ cool-down starts at cue end
      $t_c^{\text{above}} \leftarrow \varnothing$
    end if
  end for
end while

Appendix B: Design changes made during implementation

Building the software exposed places where the first design contradicted itself or the hardware. Each one is now settled in Chapter 4, and Table 8.1 records them. The class, entity-relationship and sequence diagrams of Chapter 4 include these changes.

Table 8.1: Changes made during implementation

Item	Problem in the first design	Decision
Cool-down	Appendix A started it at the first alert, which blocked Tiers 2 and 3	It starts when the cue ends, as in the state diagram
Light colours	One MOSFET can only switch the bar, not change its colour	Each class keeps its flash pattern, and colour shows on the display
Voice messages	“You seem tired” named no class, count or duration, failing T-16	Every message names the class and the count or duration
Speed tip	“Mostly at speed” needs a vehicle signal we do not have	The tip counts repeat events across trips
Event table	One row per tier would count a cue three times, and T-06 wants muted cues logged	Added an `escalated` column, muted cues use tier 0
Location	FS-26 said GPS, while Chapter 4 used the phone	Phone location, and browsers only allow it over HTTPS, so the Pi needs a certificate
Dashboard login	Uploading a 4-digit PIN hash would be weak	PINs stay in the car, and the dashboard has its own password
Power-cut claim	`synchronous=NORMAL` does not make every commit survive a power cut	Text corrected, value kept until the relay test decides
Eye closure	75% EAR fired about 100 times per hour	50% EAR, blink score and head pitch (Section 6.3.3), adding a `pitch_rest` column

Appendix C: Submitted proposal

University of Sharjah
College of Computing and Informatics
Computer Engineering Department
Senior Design Project
(Please Modify ALL RED colored text)

Project Number	Project Title : **Design and Implementation of an Edge Based Driver Monitoring System with False Alarm Reduction**
SDP1-34	**العنوان باللغة العربية تصميم وتنفيذ نظام مراقبة للسائق قائم على الحوسبة الطرفية مع الحد من الإنذارات الخاطئة**

Term: Fall **Year: 2026-2027**

Students Details:

Name	**ID**	**Major**
OUMAR MAMOUN IBRAHIM	U22200741	**CPE**
ABDULLAH JAFAR ABDELHAMID	U23101327	**CPE**
FARIS MUNADHIL AHMED ALAGHA	U23104246	**CPE**
MOHAMED AHMED SAAD KHAFAGY	U20101649	**CPE**

Project Specialty (PLEASE TICK UP TO TWO CHOICES ONLY):

CyberSecurity/Software	
Hardware/embedded systems	X
AI/Robotics	X
General Computer	

Supervisor(s) Details

Full Name: Mohamad Khairi Bin Ishak Signature: Signed by email? (**YES**/NO)

1- **Problem statement** (Maximum 150 words)

As countries increasingly mandate the usage of in-cabin Driver Monitoring Systems (DMS) [1] aiming to minimize accidents caused by driver drowsiness and distraction, edge deployment of lightweight vision models has become essential. However, edge system deployment faces three core bottlenecks:

1. False detections can trigger unnecessary alerts and downstream operations, potentially reducing driver trust and system reliability.
2. Real time monitoring requires low inference latency, while limited computational resources, memory, power, and thermal capacity restrict the use of computationally intensive models on edge devices.
3. Variations in cabin conditions and visible (RGB) and Near Infrared (NIR) illumination can affect model performance and increase susceptibility to false detections.

This introduces a clear tradeoff: Higher compute systems effectively minimize false detections, yet their footprint renders them infeasible in real-time deployment.

Overall, this project addresses this challenge through lightweight object detection, hard negative mining, and embedded system implementation.

2- **Briefly describe the project summary, importance and Objectives**

(Maximum 150 words)

Drowsiness and distraction are major factors of road crashes and contribute to more than 35% of road fatalities [2]. While in-cabin Driver Monitoring Systems (DMS) help to reduce these risks, these systems frequently produce nuisance false alarms, which undermine driver trust, encouraging the driver to circumvent the monitoring system[3].

The project has three main objectives:
(1) to evaluate lightweight object detection models using RGB and NIR driver monitoring datasets
(2) to investigate negative sample ratios and hard negative mining for reducing false detections while maintaining detection performance
(3) to deploy the selected model on a Raspberry Pi based prototype integrating a camera and alert mechanism. The final system will be evaluated using detection accuracy, F1 score, false detections per 1,000 negative frames, inference latency, and frame rate. The project combines AI model development, dataset optimization, embedded implementation, and real time system evaluation.

3- **Briefly describe the project methodology** (Maximum 150 words)

We follow a three-phase engineering methodology to realize the edge DMS. First, we benchmark nano-scale (<5M parameters) detection families, such as DFINE and YOLO, across RGB and NIR in-cabin 30FPS video datasets such as (DMD and Drive&Act). Models are evaluated on localized (not temporal) inattention cues such as (yawning, eating, drinking) using an equally-weighted normalized score (mAP:50, macro-F1, FPS, false detections per 1,000 frames) to render the optimal model that balances accuracy, detection and latency. Second, we calibrate the dataset across negative-to-positive frame ratios such as (0%,20%,40%,60%) and curate the dataset with hard false positives instead of random negative frames to suppress false detections.

Finally, the selected lightweight model will be optimized and deployed on a Raspberry Pi based edge platform integrated with an RGB camera and real time alert mechanism. The complete embedded system will be experimentally evaluated in terms of inference speed, processing latency, detection reliability, false alarm rate, and overall real time performance under representative driver monitoring conditions.

4- **Briefly describe the expected outcomes** (Tick one or more of the following choices)

What are the outcomes of your project: e.g. software, prototype that does so and so, simulation to prove research points, conference paper, etc. Write a paragraph and then select from the table below.

Outcome	Choice
Physical Prototype	**X**
Simulation demo	
Research paper	**X**
Software app/module	

The project delivers three main outcomes. Firstly, a report covering the literature review, dataset curation methodology and embedded deployment results. Second, an optimized lightweight object detection model for edge-based Driver Monitoring Systems, trained using curated data to reduce false detections while maintaining reliable detection performance. Third, a functional embedded prototype implemented on a Raspberry Pi 4, integrating a real time RGB camera, edge AI inference, and an alert mechanism for in cabin driver monitoring. The prototype will demonstrate the practical feasibility of deploying lightweight AI models for reliable and real time driver monitoring on resource constrained edge hardware.

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